A

Project Report on

IDENTIFICATION OF PARKINSON'S DISEASE BY VOICE AND SPIRAL DRAWINGS USING MACHINE LEARNING

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CERTIFICATE

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With Regards,

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ABSTRACT

Parkinson’s disease (PD) is a neurodegenerative movement disease where the symptoms gradually develop start with a slight tremor in one hand and a feeling of stiffness in the body and it became worse over time. It affects over 6 million people worldwide. At present there is no conclusive result for this disease by non-specialist clinicians, particularly in the early stage of the disease where identification of the symptoms are very difficult in its earlier stages. The proposed predictive analytics framework is a combination of K-means clustering and Decision Tree which is used to gain insights from patients. By using machine learning techniques, the problem can be solved with minimal error rate. In this model, a huge amount of data is collected from the normal person and also previously affected person by Parkinson's disease, these data is trained using machine learning algorithms. From the whole data 60% is used for training and 40% is used for testing. There are 24 columns in the data set each column will indicate the symptom values of a patient except the status column. The status column has 0's and 1’s. Those values will decide the person is affected with Parkinson’s disease. 1's indicate person is affected, 0's indicate normal conditions. Also, OpenCV (Open Source Computer Vision Library) a library of programming functions mainly aimed at real-time computer vision was built to provide an infrastructure for computer vision applications and to accelerate the use of machine perception in the real time. Thus, our output will showcase the early detection of the disease and can be able to increase the lifespan of the diseased patient with proper treatments and medications leads to peaceful life.

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CHAPTER-1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Parkinson's disease is the world’s second-ranking common brain disease which affects brainpower and results in unintended body movements. Early signs of Parkinson's disease are very much difficult to notice. The main cause of Parkinson’s disease is the deficiency of the hormone dopamine that is present in the brain. It is produced by brain cells. it acts as a message-bearer between brain neurons and other body parts. The loss of dopamine affects the transmission of messages between the brain and other body parts.

Main reasons for causing Parkinson’s Disease are:

- **Age:** Parkinson's disease is a rare occurrence among young adults. It usually starts in middle or late life, and the risk gets higher as you get older. The disease typically strikes people at 60 or older. Making family planning decisions may be aided by genetic counselling if a young person is diagnosed with Parkinson's disease. Also distinct from those of an older individual with Parkinson's disease and requiring special consideration are work, social circumstances, and drug side effects.

- **Heredity:** The likelihood that you'll develop Parkinson's disease increases if you have close family members who have the condition. Unless you have a large number of family members who suffer from Parkinson's disease, your risks are still minimal. and adverse medication reactions.

- **Head Injuries:** Damaging of brain nerves due to head injuries because off several accidents is one of the main reasons for Parkinson’s disease

- **The presence of Lewy bodies**: Microscopical indicators of Parkinson's disease include clumps of particular chemicals within brain cells. Lewy bodies are what they are, and scientists think they offer a crucial insight into what causes Parkinson's disease.

Formation of tremors is usual in human beings. But some tremors are unnoticeable in early stages but becoming dangerous gradually. PD patients are attacked by one of the tremors, which causes to cumulative sickness and affects the nerves on time being. The tremors, which lead to PD, develop barely in one hand of the person, which may be inconspicuous at early stages. But they will be slowing the movements of limbs such as legs, hands, thumbs and develops stiffness in them. The facial expressions of the patient of PD will be downing gradually at early stages and they are not able to swing their arms while walking. Speech become mumble or slurred [2].

The symptoms of this disease become terrible over time. Despite the fact that there the PD can't be cured but medication helps in reducing the effect of symptoms significantly, if the disease identified in an early stages. Sometimes, doctors may even advise surgery to regulate certain regions of the brain [3]. Person affected by this disease May slowing lose the ability to give facial expressions as a normal person and may not be able to wave their hands while moving. They will speck indistinctly. The condition of the person may become dreadful over a period of time. In spite of that this sickness is incurable but early detection the disease helps in reducing the effect of this sickness. In some cases surgeries are also options to regulate some parts of brain [3].

At first limb experience the tremor, which is rhythmic shaking, typically the hand or fingers. That could wiggle your thumb and forefinger. Even when diseased persons are at ideal state, the hand may shake. As the patient work, the shaking might be less. PD may slow your movements over time, which makes challenging and long delayed. Changes in the overall health can also impact PD symptoms [3].

Some of the health concerns can be linked to an increase in symptoms or worsening of the disease like infections and medication changes. Eventually they are not able to take longer steps while walking. It may become difficult to get up from a chair. Firmness in muscles may occur in the body. This will cause lot of pain and the person will not be able to do the work or move as usual. This disease also affects the posture of the body, slowly lose the balance and fall down, might not be able to perform insentient movements like blinking, specking, smiling or swinging arms, specking quickly and eventually speech may become soft and slurry [4].

Symptoms will differ from individual. Sometimes these symptoms are barely noticeable in the starting stages of this disease. Even though symptoms get to affect the legs on either side of the body, they usually start affecting some position of the body and continue to become more complex there. The most common signs of this disease are tremors, slowed movements, inflexible muscles, weak posture and balance of the body, unable to perform unconscious movements, face difficulty in writing, irregularities in speech [4].

Fig 1.1 Symptoms of Parkinson's Disease

The effect of PD will be different from one person to another. The symptoms of this disease are unnoticeable in the beginning. Tremor affects the lower limbs of one side the body and affects the posture of the body and become more vicious over time. The usual symptoms of this disease are reducing the speed of gestures, stubborn muscles, and frail body position and may not be able to do unconscious gestures and may become hard in writing and speaking. The posture of the body should be the main consideration when predicting the disease [5].

Parkinson's has four main symptoms:

- Tremor in hands, arms, legs, jaw, or head.
- Muscle stiffness, where muscle remains contracted for a long time.
- Slowness of movement.
- Impaired balance and coordination, sometimes leading to falls.

There are 5 stages of Parkinson's disease

Stage 1: Does not affect the daily routines of a person

Stage 2: Starts affecting the routine activities.

Stage 3: Takes more time to complete the routine works.

Stage 4: Difficult to do their own activities.

Stage 5: Confined to bed and the person will die in a short period of time.

The most common problems caused by Parkinson's disease are

Tremor:

The hand or fingers are frequently the first part of the body to tremble or shake rhythmically. Your thumb and forefinger can be moved. This is referred to as a "pill-rolling tremor". Your hand could tremble even when it is at rest. The shaking could reduce while you working.

Rigid body movements:

Become very hard to move body parts.

Change in voice:

We can observe changes in the voice the means there will be some deflection in voice and hard speaking in advanced stages.

Development of quivering is common with increasing age but some tremors are inconspicuous in the early stage but may lead to vicious consequences. Tremors developed in the patients affected by PD cause aggregative sickness in the person affecting the nervous system. This kind of tremors will affect the movement of the limbs such as hands, legs and thumbs and develop rigidity in the PD person [6]. In most cases, when PD is diagnosed, it has already developed to an advanced stage, there has been significant neuron loss and damage, at that stage any treatment will be may not be able to stop the disease from spreading further or lead to neuroprotection. The main goal of this study is to identify the characteristics of limb movements of the person affected by this disease, and by using machine learning the status of the disease can be identified with earlydetection, the person affected with disease can treated so that the person can live longer [6].

Since there is no unassailable test for PD, the prognosis of the disease must now only be made using clinical and observational criteria. Many of the symptoms of PD are difficult to identify and are alsoseen in other disorders. To assess the severity of the condition, a doctor will use the UPDRS, a measure based on a score produced from the neurological evaluation. However, because it is a qualitative perception, the scale lacks objectivity, reproducibility and sensitivity [7].

The most prevalent pre-diagnostic symptom of PD istremor, which 41% of patients report to their doctors in the preceding two to diagnosis, compared to lessthan 1% of controls. At five and ten years prior to diagnosis, the incidence of tremor are already high [7].Machine learning algorithms have shown promise in detecting PD based on various biomarkers, including spiral drawings and voice recordings[8].

Spiral drawings are a common biomarker for PD, as the tremors associated with the disease can cause irregularities in the drawing pattern. Machine learning algorithms can analyze these patterns and identify the presence of PD with high accuracy [8].Similarly, voice recordings can also be used to detect PD, as the disease can affect the vocal cords and cause changes in speech patterns. By analyzing features of the voice, such as pitch and tremors, machine learning algorithms can identify the presence of PD [9].

XGBoost is a popular machine learning algorithm for classification of tasks.. XGBoost is a gradient boosting algorithm that also uses an ensemble of decision trees but focuses on optimizing the gradient descent algorithm to minimize the loss function [9]. The DenseNet (densely connected convolutional network) is recognized for having convolutional neural network architecture that is state-of-art, when validated for classification using the popular ImageNet dataset. Huang et al. validated the technique of using direct connections in a feed-forward manner from each layer to every other layer. Every layer in the model architecture takes the target input and concatenation of the preceding layers' feature maps. A DenseNet is a type of convolutional neural network that utilises dense connections between layers, through Dense Blocks, where we connect all layers (with matching feature-map sizes) directly with each other. It performs non-linear operations such as batch normalization, ReLU, and convolution or pooling. If the size of the feature maps changes, the concatenation procedure is unsuccessful. [9].

1.2 SCOPE

PD is a neurodegenerative disorder that affects the motor system, leading to symptoms such as tremors, rigidity, and bradykinesia. Early diagnosis and treatment of PD are crucial for slowing down the progression of the disease and improving the quality of life of patients [8].

The scope behind using Densenet and XGBoost for PD detection is to create a model that can accurately diagnose the disease based on patient data. This data may include information such as age, gender, medical history, and symptoms. Spiral drawings are a common biomarker for PD, as the tremors associated with the disease can cause irregularities in the drawing pattern. Machine learning algorithms can analyze these patterns and identify the presence of PD with high accuracy [8]. By training the machine learning model on a large dataset of patient information, it can learn to identify patterns and correlations between these variables and the likelihood of PD [9].

Once the model is trained, it can be used to analyze new patient data and provide a diagnosis. Thiscan help doctors to make a more accurate diagnosis, potentially at an earlier stage of the disease, which can lead to better treatment outcomes [9].

The use of machine learning algorithms on spiral drawings and voice recordings has several advantages over traditional diagnostic methods such as neurological exams and brain imaging. These methods can be time-consuming, expensive, and invasive. Using machine learning algorithms can lead to faster and more accurate diagnosis of PD, which can help patients receive timely treatment [10].

1.3 INTRODUCTION TO MACHINE LEARNING

Machine Learning is the sub-area of AI, whereby the term refers to the power of IT systems to independently find solutions to problems by recognizing patterns in databases. Machine Learning is an AI technique that teaches computers to learn from experience. Machine learning algorithms use computational methods to "learn" information directly from data without relying on a predetermined equation as a model. In other words: Machine Learning enables IT systems to acknowledge patterns in the idea of existing algorithms and data sets and to develop adequate solution concepts. Therefore, in Machine Learning, artificial knowledge is generated on the idea of experience. While artificial intelligence and machine

learning are often used interchangeably, they are two different concepts. AI is the broader concept – machines making decisions, learning new skills, and solving problems in a similar way to humans – whereas machine learning is a subset of AI that enables intelligent systems to autonomously learn new things from data. Instead of programming machine learning algorithms to perform tasks, you can feed them examples of labeled data (known as training data), which helps them make calculations, process data, and identify patterns automatically.

In order to enable the software to independently generate solutions, the prior action of 5 people is important. For example, the required algorithms and data must be fed into the systems in advance and the respective analysis rules for the recognition of patterns in the data stock must be defined.

Once these two steps have been completed, the system can perform the following tasks by Machine Learning [10]:

- Finding, extracting and summarizing relevant data
- Making predictions based on the analysis data
- Calculating probabilities for specific results

Basically, algorithms play a crucial role in Machine Learning: On the one hand, they're liable for recognizing patterns and on the opposite hand, they will generate solutions. Algorithms can be divided into different categories [11].

1.3.1 Supervised Learning

Techniques of Supervised Machine Learning algorithms include linear and logistic regression, multi-class classification, Decision Tree, and Support Vector Machine. Supervised Learning problems area kind of machine learning technique often further grouped into Regression and Classification problems. The difference between these two is that the dependent attribute is numerical for regression and categorical for classification [11].

- **Regression :** Linear regression could also be a linear model, e.g. a model that assumes a linear relationship between the input variables (x) and thus the only output variable (y). More specifically, that y is usually calculated from a linear combination of the input variables (x). Whenthere's one input variable (x), the tactic is mentioned as simple

linear regression. When there are multiple input variables, literature from statistics often refers to the tactic as multiple linear regression [11].

- **Classification :** Classification could also be a process of categorizing a given set of data into classes, It is often performed on both structured or unstructured data. the tactic starts withpredicting the category of given data points. The classes are often mentioned as target, label, or categories. In short, classification either predicts categorical class labels or classification data supported the training set and thus the values(class labels) in classifying attributes and uses it in classifying new data. There is a variety of classification models. Classification models include Logistic Regression, Decision Tree, Random Forest, Gradient Boosted Tree, One-vs.-One, and Naïve Bayes [11].

1.3.2 Unsupervised Learning

There In unsupervised learning, AI learns without predefined target values and without rewards. It's mainly used for learning segmentation (clustering). The machine tries to structure and type the info entered consistent with certain characteristics. Unsupervised Learning is that the training of Machines using information that's neither classified nor labeled and allowing the algorithm to act there on information without guidance.

Unsupervised Learning is accessed into two categories of algorithms [11]:

- **Clustering :** A clustering problem is where you would like to get the inherent grouping in the data such as grouping customers by purchasing behavior.
- **Association :** An Association rule learning problem is where it would wish to get rules that describe large portions of your data such as folks that buy X also tend to shop for Y.

1.3.3 Classification Metrics

Classification metric is a number that measures the performance that your machine learning model when it comes to assigning observations to certain classes. Binary classification is a particular situation where you just have to classes both positive and negative. Classification is a supervised learning task in which it will try to predict the class or label of a data point based on some feature values.Depending on the number of

classes target variable includes, it can a binary or multi-class classification. Evaluating a machine learning model is just as important as building it.

Classification measurements are the type of measurements which establish the a priori ill-defined composition and essence of the designations (classes) of a categorical measurement scale and the result is the assignment of a single object to one of the chosen classes. The most straight forward way to measure a classifier's performance is using the Accuracy metric. Here, we compare the actual and predicted class of each data point, and each match counts for one correct prediction. Accuracy is then given as the number of correct predictions divided by the total number of predictions. Evaluation metrics are tied to machine learning tasks. Using different metrics for performance evaluation, we should be able to improve our model's overall predictive power before we roll it out for production on unseen data. Without doing a proper evaluation of the Machine Learning model by using different evaluation metrics, and only depending on accuracy, can lead to a problem when the respective model is deployed on unseen data and may end in poor predictions. classification metric is a number that measures the performance that your machine learning model when it comes to assigning observations to certain classes. Binary classification is a particular situation where you just have to classes: positive and negative. The optimal choice of metric usually depends on the characteristic of the data and the given task. It will also explain when a particularmetric might be a better fit than the others [12].

1. **True Positives :** It is the case where we predicted Yes and the output is also Yes.

2. **True Negatives :** It is the case where we predicted No and the output is also No.

3. **False Positives :** It is the case where we predicted Yes but, it is actually No.

4. **False Negatives :** It is the case where we predicted No but, it is actually Yes.

The various machine learning classification metrics for different algorithms are given below:

Accuracy: One parameter for evaluating classification models is accuracy. Accuracy is defined as the percentage of correct predictions for the test data. It can be calculated easily by dividing the number of correct predictions by the number of total predictions. The percentage of predictions that the model correctly predicted is known as accuracy. The following is the actual definition of accuracy [12].

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

For binary classification, accuracy can also be calculated in terms of positives and negatives as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Sensitivity: It is a measure of the proportion of actual positive cases that got predicted as positive (or true positive). Sensitivity is used to evaluate model performance because it allows us to see how many positive instances the model was able to correctly identify. This implies that there will be another proportion of actual positive cases, which would get predicted incorrectly as negative (or false negative) [12].

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Specificity: It is defined as the proportion of actual negatives, which got predicted as the negative (or true negative). This implies that there will be another proportion of actual negative, which got predicted as positive and could be termed as false positives. This proportion could also be called a false positive rate [12].

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Precision: It is the Ratio of true positives to total predicted positives.

$$\textbf{Precision} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FP}}$$

Recall: It is calculated as the proportion of Positive samples that have been accurately identifiedas Positive to all Positive samples. The recall evaluates how well the model can identify positive samples. The more positive samples that are identified, the greater the recall [12].

$$\textbf{Recall} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FN}}$$

F1-Score: It is a harmonic mean between recall and precision. Its range is [0,1 This metric typically indicates how precise (how many times it correctly classifies) and how robust the system is. It is used to measure the test's accuracy [12]**.**

$$\textbf{F1-Score} = \frac{\textbf{2 * Precision * Recall}}{\textbf{Precision} + \textbf{Recall}}$$

CHAPTER-2

LITERATURE SURVEY & EXISTING SYSTEM

2.1 INTRODUCTION

Machine learning techniques have shown promise in the early detection of PD. In recent years, several studies have been conducted to develop machine learning models for the detection of PD using various types of data, such as voice, gait, and handwriting, among others.

In [7], Aleksandr Talitcki et al target the recognition of exercises enabling precise PD detection. The author created a collection of 15 tasks to help individuals perceive the symptoms that are uniquely linked to PD.The exercises are grouped into 4 classes they are Gross Motor (GM), Clinical Evaluation (CE), Fine Motor (FM) and Tremor at Rest. The first of the Gross Motor (GM) is used to measure the subject's movements, which are frequently employed in daily activities. The clinical evaluation (CE) entails customary techniques that neurologists employ to identify the tremor. For the purpose of evaluating the subject's fine alignment, fine motor (FM) tests are required. While the body is restrained, Tremor at Rest is employed to help expose the tremor.

Each measurement was carried out at 100 Hz and transmitted through BLE to an Intel Next Unit of Computing (NUC) computer that was running by the specialised Python software. Although it is advised, applying the entire exercises can be disruption and long delayed for both the patient and the practitioner. The primary limitations of this study are wireless communication and the possibility that people with the fourth stage of PD might find it difficult to wear even the smallest sensing device. The wireless transmitter is also the device on board that consumes the more current, and the battery can run out in about an hour.

In [1], Juan C. Pérez-Ibarra et al proposed two online causal algorithms for gait event detection and phase segmentation. A simple strategy is to split the gait into two things: Position and Motion. The author suggestion is to alternate between various linear classifiers, a single one for each stage, enabling the classifier changes whenever the FSM transitions to a new state. The author employed a mix of the Simulated Annealing (SA) and Genetic meta-heuristics techniques (GA). The author has created a dyad of wearable gait

analysis gadgets. They were created to be worn over the subjects' normal footwear for regular activities. They created an upgraded model of our wearable IMU-based gadget that records and measures gait information.

In [13], Laiba Zahid et al proposed move learning based procedures and AI based procedure is used. In AI based procedure utilized help vector machine (svm) and arbitrary woodland calculation is utilized. The most elevated 99.7 percent exactness on vowel \o\ and read text is noticed utilizing a multifacet discernment. While 99.1 percent exactness saw on vowel \i\ profound elements utilizing irregular backwoods. The proposed procedure beats the current methods on the pc-Gita dataset for Parkinson's infection location.

In [14], Liaqat Ali et al said to dealt with two significant problems, first issue biasedness by imbalanced informational collection for that they utilized irregular undersampling technique using machine learning. What's more, second issue is low pace of arrangement accuracy, to further develop it they utilized flowed learning framework method.it was likewise seen that the proposed flowed framework works on the solidity of traditional has 3.3 percent by Adaboost model and decreases its intricacy. Moreover, the flowed framework accomplished characterization precision of 76.44%, awareness of 70.94% and explicitness of 81.94%. Restrictions are Not more exact contrasted with different calculations.

In [15], Federica Amato et al separated 60 elements from previous handled vocal signals and involved them as contribution to a few AI models. 92 % exactness in 10-crease cross-approval utilizing SVM. Limitations are Just the SVM calculation works among all the diff calculation utilized in the undertaking. Results show that the 7 elements chose for SVM accomplishes great generally speaking exactness of 83.33%, great location rate for Parkinson`s infection of 75% and low bogus positive after effects of 16.67%. There is a model for recognizing Parkinson's utilizing voice. The diversions in the voice will affirm the side effects of Parkinson's sickness. This task showed 73.8% effectiveness. In our model, a colossal measure of information is gathered from the typical individual and furthermore recently impacted individual by Parkinson's sickness. There is a model for recognizing Parkinson's utilizing voice. The diversions in the voice will affirm the side effects of Parkinson's sickness. This task showed 73.8% effectiveness. In our model, a colossal measure of information is gathered from the typical individual and furthermore recently impacted individual by Parkinson's sickness.

In [16], N. P. Narendra et al proposed two approaches to recognize the disease first approach is standard pipeline approach in that they used three methodology, they are system structure, baseline features and gif strategies and glottal features. Second approach is beginning to end approach called Pipeline approach. From the ordinary pipeline structures, the most raised grouping accuracy (67.93%) was given by blend of benchmark what's more, QCP-based glottal components. From the very outset to complete systems, the most raised accuracy (68.56%) was given by the structure arranged using QCPbased glottal stream signals. Regardless of the way that gathering exactnesses were honest for all systems, the audit is engaging as the extraction of voice source information was seen as most convincing in the two strategies.

In [17], K. Polat proposed the cross variety approach to investigate Parkinson's contamination using talk signals. It combines resounding and time-repeat parts for strong component extraction. Ability of Convolutional Mind Associations for voice signal assessment is researched. The author used two strategies they are SMOTE(Synthetic Minority Over-testing Strategy) and sporadic forest area. SMOTE approach, the amount of tests for minority class in the PD dataset has been falsely extended to change the dataset.

Precision of simply irregular backwoods is 87.037% combination of both Destroyed and irregular timberland precision is 94.89%. Accuracy is the degree of closeness between a measurement and its true value. Precision is the degree to which repeated measurements under the same conditions show the same results. The sporadic forest area technique achieved a higher responsiveness than the decision tree model. Subsequently, it is fitting to cultivate a show to helpfully recognize starting stage PDD considering the PD-MCI gauge model made in this survey, to spread out individualized seeing to follow high-risk social occasions. People with Parkinson's disease may lose their sense of smell and experience sleep disruptions during the rapid eye movement sleep period. It is estimated that 1% of the population over the age of 60 suffers with paralysis against. . Parkinson's disease symptoms vary from person to person. Early warning indicators are subtle and go overlooked. Symptoms normally start on one side of your body and progress to the opposite side before affecting both edges.

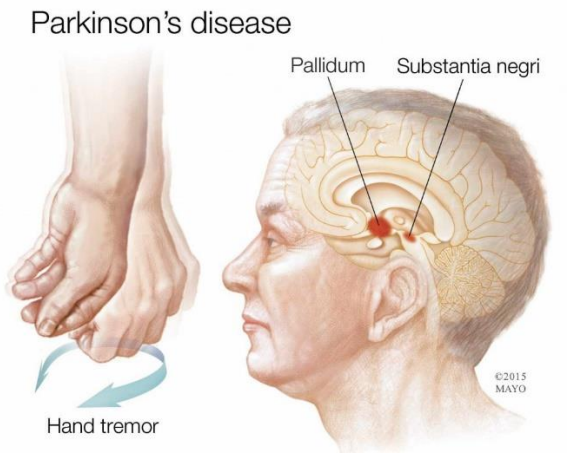

Fig 2.1 Hand tremors caused by Parkinson’s Disease

2.2 EXISTING SYSTEM & DRAWBACKS

In PD can only be recognized at secondary stages under the current system, making disease prevention is challenging and potentially fatal. In this system, the doctor must manually assess the patient's symptoms by taking the MRI scans which describes the nigral structure of patients to identify the disease. Each person has a unique set of sickness symptoms. Based on the brain condition and dopamine levels, doctors with special training in brain disorders can identify the disease [5]. Doctors typically use MRIs, SPECT scans, and blood tests. They are also known as image tests. In order to create 3D images of the brain that allows the doctor to understand the status of the brain by the SPECT which uses nuclear radio power. To provide an image of the organs, MRI uses radio frequency bands and magnetic flux [5]. There are various systems for the detection of Parkinson's disease, ranging from clinical assessment tools to computer-based diagnostic methods. Some of the most commonly used methods are:

1) **Clinical Assessment:** The most common method for diagnosing Parkinson's disease is through clinical assessment by a neurologist. The neurologist will evaluate the patient's medical history, physical examination, and symptoms to determine if they meet the criteria for a diagnosis of Parkinson's disease.

2) **Rating Scales:** There are several rating scales available that can be used to assess the severity of Parkinson's disease symptoms. These scales include the Unified Parkinson's Disease Rating Scale (UPDRS) and the Hoehn and Yahr Scale.

i. **Unified Parkinson's Disease Rating Scale (UPDRS):** This is a clinical tool used to evaluate the severity of Parkinson's disease symptoms. It consists of four parts: (1) mentation, behavior, and mood, (2) activities of daily living, (3) motor examination, and (4) complications of therapy.

ii. **Hoehn and Yahr Scale:** This is another clinical tool used to assess the severity of Parkinson's disease based on the degree of disability and motor impairment. It consists of five stages, ranging from mild symptoms on one side of the body to severe disability on both sides.

3) **Montreal Cognitive Assessment (MoCA):** This is a cognitive screening tool used to detect mild cognitive impairment in patients with Parkinson's disease. It consists of various tasks such as memory, attention, language, and visuospatial abilities.

4) **Computer-based methods:** Various computer-based methods have been developed for the detection of Parkinson's disease, such as machine learning , deep learning models, and neural networks. These methods use data from various sources such as voice recordings, gait analysis, and eye movements to detect Parkinson's disease with high accuracy.

5) **Imaging techniques:** Imaging techniques such as positron emission tomography (PET) and magnetic resonance imaging (MRI) have been used to detect changes in the brain that are associated with Parkinson's disease. These techniques can detect changes in the dopamine system and other brain structures affected by the disease.

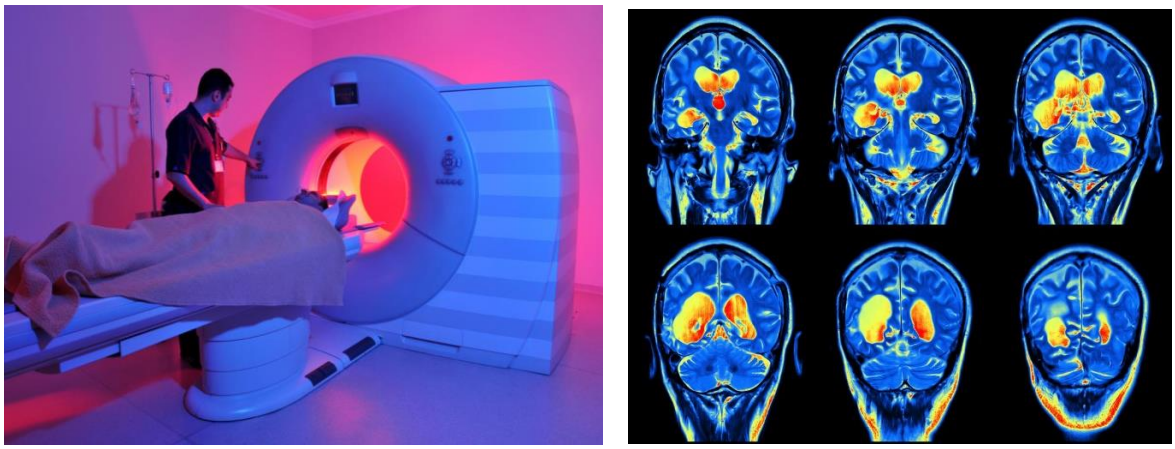

Fig 2.2 MRI Brain Scans

6) **Electrophysiological Techniques:** Electrophysiological techniques such as electroencephalography (EEG) and electromyography (EMG) can be used to measure brain activity and muscle activity, respectively, to help diagnose Parkinson's disease.

7) **Blood Tests:** Blood tests can be used to detect specific biomarkers associated with Parkinson's disease, such as alpha-synuclein and DJ-1.

8) **Speech Analysis:** Changes in speech patterns can be an early sign of Parkinson's disease, and speech analysis tools can be used to detect changes in speech patterns that may be indicative of the disease.

2.2.1 Drawbacks

There are several limitations of existing systems for the detection of Parkinson's disease, including:

1) **Lack of accuracy:** Existing systems for Parkinson's disease detection may not be accurate enough to provide reliable results. This can be due to factors such as variability in symptoms, differences in disease progression among patients, and the sensitivity of the detection method.
2) **Limited availability:** Many existing systems for Parkinson's disease detection are only available in specialized clinics or research centers, making them inaccessible to a large number of patients.
3) **Costly:** Some systems for Parkinson's disease detection can be costly, making them prohibitive for patients who cannot afford the expense.
4) **Lack of standardization:** There is currently no standardization in the way that Parkinson's disease is diagnosed and detected, which can lead to inconsistencies in the results obtained from different systems.
5) **Limited scope:** Some existing systems for Parkinson's disease detection focus only on certain symptoms or aspects of the disease, which may not provide a comprehensive picture of the patient's condition.
6) **Invasive methods:** Some existing systems for Parkinson's disease detection require invasive procedures such as lumbar punctures or brain imaging, which can be uncomfortable and risky for patients.

7) **Subjectivity of Diagnosis:** Currently, the diagnosis of PD relies heavily on clinical assessments and subjective evaluations by neurologists, which can lead to inconsistencies and inaccuracies in diagnosis.
8) **Lack of Early Detection:** PD symptoms often do not appear until significant damage has already occurred in the brain, which can limit the effectiveness of treatments and interventions.
9) **Lack of Personalization**: Current diagnostic tests do not take into account individual differences in disease progression or response to treatment, which limits the ability to personalize treatment plans for PD patients.

2.3 PROPOSED SYSTEM

The proposed system for the detection of Parkinson's disease using Random Forest and XGBoost would involve the use of spiral drawings and voice deflections as input data for the model. Firstly, the patient would be asked to draw a spiral on a piece of paper using a pen or pencil. The drawing would then be scanned and processed to extract features such as the number of turns, the length of each turn, and the amount of tremors during the drawing. These features would then be fed into the Random Forest and XGBoost models to predict the likelihood of the patient having Parkinson's disease.

In addition to spiral drawings, the patient's voice would also be recorded and analyzed for signs of Parkinson's disease. Features such as tremors, pitch, and volume would be extracted from the voice recording and used as additional input for the models.

The XGBoost model would be trained on a large dataset of spiral drawings and voice recordings from both Parkinson's disease patients and healthy individuals. The models would then be used to predict whether a new patient is likely to have Parkinson's disease based on their spiral drawing and voice recordings.

To develop a machine learning model for PD detection using voice recordings, a dataset of voice recordings from both healthy individuals and those with PD would be needed. The dataset would need to be labelled to indicate which recordings were from individuals with PD and which were from healthy individuals [18].

The next step would be to pre-process the data, which might involve feature extraction to identify relevant characteristics of the voice recordings that could be used as inputs to the

machine learning model. These features could include measures of pitch, loudness, and other acoustic properties of speech [18]. Once the data has been pre-processed, it could be used to train a machine learning model using techniques such as logistic regression, decision trees, or neural networks. The model would be trained to recognize patterns in the voice recordings that are indicative of PD [19].

Finally, the trained model could be tested on a separate dataset of voice recordings to evaluate its performance. This evaluation might involve metrics such as accuracy, precision, recall, and F1 score to assess how well the model is able to correctly identify individuals with PD based on their speech patterns.

The proposed system overcomes the limitations of the previous models with the high accuracy Spiral data and speech datasets are both inputs for this system. To measure the deflections in the voice, linear regression and XGBoost is applied to the voice dataset. Due to a variety of factors, like the inability to hold a pen properly or fear, not everyone is able to draw the spiral shape. The spiral model's issues are resolved by the voice model. The system is hence highly accurate [19]. The massive amount of spiral and audio data from healthy and ill patients are collected. Training and testing sets were derived from these datasets. When used separately, spiral and voice models are unable to produce accurate outcome. So, each model has its own drawbacks. This combines the results of the spiral and voice models to identify the illness [20].

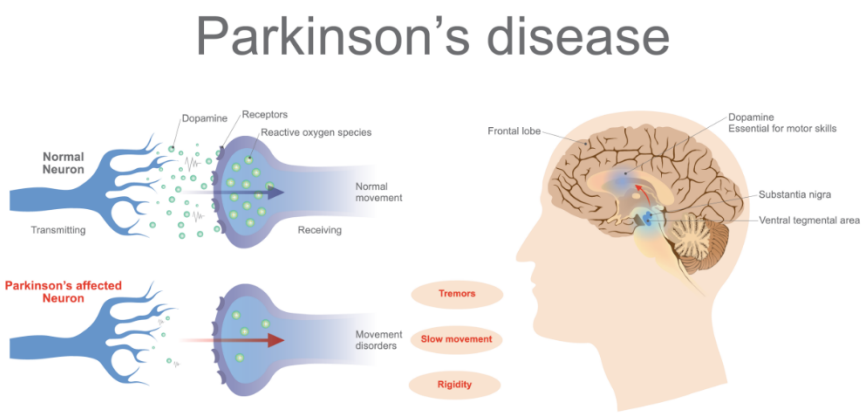

Fig 2.3 Parkinson's Disease Affected and Not Affected neurons

2.4 FEASIBILITY STUDY

The machine learning feasibility study phase is used to dig into the data and quickly conduct experiments to establish baseline performance on a task. Some key considerations when conducting a feasibility study for detecting PD using machine learning are given below [20]:

- **Data collection:** Voice and spiral datasets were collected kaggle repository.
- **Data preprocessing:** Once the data has been collected, it should be preprocessed. Preprocessing involves cleaning and transforming the data into a suitable format for machine learning algorithms. It involves removing missing values, outlier detection, normalization, and feature selection.
- **Machine learning algorithms:** Machine learning algorithms such as Random Forest and XGBoost to detect PD. Random forest is applied on spiral dataset and XGboost was applied on voice dataset. These algorithms are powerful and have been proven effective in detecting complex patterns in large datasets.
- **Model evaluation:** We can evaluate the performance of the machine learning model to determine its accuracy in detecting PD. The model should be evaluated using various metrics such as accuracy, confusion matrix.
- **Clinical validation:** The final step in conducting a feasibility study is to clinically validate the machine learning model. Clinical validation involves testing the model on a separate dataset to determine its generalizability and accuracy in real-world settings.

2.4.1 Economical Feasibility

The economic feasibility of using machine learning algorithms like Random Forest and XGBoost for detecting PD using spiral drawings and voice deflections can be analyzed from various perspectives [17]:

- **Cost of data collection:** Collecting data for training and testing the machine learning models can be expensive. In the case of PD detection, the cost of collecting spiral drawings and voice deflections from patients can be high. However, this cost can be reduced by using digital platforms that allow remote data collection and sharing.

- **Cost of model development:** Developing machine learning models requires significant expertise and resources. Hiring data scientists, machine learning engineers, and software developers can be expensive. However, there are open-source tools and libraries available that can be used to build models without incurring significant costs.
- **Accuracy of the model:** The accuracy of the machine learning models is crucial in determining their economic feasibility. If the models are not accurate enough, then they will not be useful in clinical settings. The Random Forest and XGBoost algorithms have shown promising results in detecting PD using spiral drawings and voice deflections, with accuracies of up to 90%.
- **Cost of deployment:** Deploying the machine learning models in clinical settings can be expensive. The cost of hardware, software, and maintenance needs to be considered. However, the cost can be reduced by using cloud-based solutions that allow remote deployment and maintenance.
- **Economic benefits:** The economic benefits of using machine learning models for detecting PD are significant. Early detection and treatment can reduce the progression of the disease, improve the quality of life of patients, and reduce healthcare costs. In addition, machine learning models can help in identifying patients who are at high risk of developing PD, enabling early intervention and prevention.

The economic feasibility of using machine learning algorithms like Random Forest and XGBoost for detecting PD using spiral drawings and voice deflections depends on several factors, including the cost of data collection, model development, accuracy of the model, cost of deployment, and economic benefits. However, the potential benefits of early detection and treatment of PD outweigh the costs, making it a promising area for research and development.

2.4.2 Technical Feasibility

The Technical Feasibility behind the detection of PD using machine learning techniques such as XGBoost using spiral drawings and voice deflections is well established. PD affects the central nervous system, which controls the movement of the body. One of the key symptoms of PD is tremors, which can be observed in the patient's handwriting and voice[18].

Spiral drawings can be used to analyze the patient's handwriting, and voice recordings can be analyzed to detect changes in speech patterns. Machine learning algorithms such as Random Forest and XGBoost can be trained on large datasets of spiral drawings and voice recordings to identify patterns that are indicative of PD. These algorithms can then be used to classify new data as either Parkinson's positive or negative. The accuracy of the classification model can be improved by including additional features such as age, gender, and other medical history [21].

Overall, the technical feasibility of using machine learning algorithms to detect PD using spiral drawings and voice deflections is well established, and this approach has the potential to improve the accuracy and speed of diagnosis for PD, which will help in the getting more correct predictions of the disease. The technical feasibility is going to give the better results for the diagnosis of the disease [22].

2.4.3 Social Feasibility

The Social Feasibility of using machine learning models such as XGBoost for the detection of PD using spiral drawings and voice deflections depends on several factors includes the following [23]:

- **Accessibility:** The availability of technology and resources required for the detection of PD using machine learning models may vary across different regions and populations. Therefore, it is important to ensure that the technology is accessible to all communities, regardless of their socioeconomic status or geographic location.
- **Acceptance:** The use of machine learning models for medical diagnosis may not be readily accepted by some individuals, particularly those who may not trust technology or may prefer traditional methods of diagnosis. Therefore, it is important to educate and inform the public about the benefits and limitations of using machine learning models for medical diagnosis.
- **Privacy:** The use of machine learning models for medical diagnosis raises concerns about privacy and data security. It is important to ensure that patient data is protected and used only for the intended purpose of medical diagnosis, and that patients are informed about how their data is being used.

- **Accuracy:** The accuracy of machine learning models for the detection of PD using spiral drawings and voice deflections is critical to their social feasibility. It is important to validate the accuracy of these models across different populations to ensure that they are effective and reliable.

Proposed projects are beneficial only if they can be turned out into information system. That will meet the organization's operating requirements. Operational feasibility aspects of the project are to be taken as a crucial a part of the project implementation. The important issues raised are to check the operational feasibility of a project includes support, application specific and management issues [24]. Overall, the social feasibility of using machine learning models for the detection of PD using spiral drawings and voice deflections depends on their accessibility, acceptance, privacy, and accuracy. By addressing these factors, we can ensure that these models are effective, reliable, and socially acceptable.

2.5 DATASET

2.5.1 Voice Dataset

Voice dataset is collected from Kaggle [25]. This contains 195 persons containing 147 `Parkinson, 48 healthy people data. This dataset contains 24 features. They are

- **MDVP:Fo(Hz) :** Average fundamental frequency of the voice. The average of all the fundamental frequency values that were derived from different periods. Areas where voices break apart are ignored.
- **MDVP:Fhi(Hz) :** Maximum fundamental vocal frequency. the highest value for fundamental frequency among all retrieved period-to-period. Areas with voice breaks are not included. The pitch extraction range is set to either search for periods from 200 to 1000 Hz or from 70 to 625 Hz. As a result, determining a fundamental above 625 Hz falls outside the "normal" range.
- **MDVP:Flo(Hz) :** Minimum fundamental frequency for voice. the lowest of all extracted fundamental frequency values for period-to-period. Areas where voices break up are excluded. The lowest fundamental for the specified time period is taken out and presented in the form of Flo. The pitch extraction range, however, is set to either search for periods between 200 and 1000 Hz or between 70 and 625 Hz.

- **MDVP:Jitter(%) :** Relative evaluation of the period-to-period (very short-term) variability of the pitch within the analyzed voice sample. Voice break areas are excluded. The pitch of the voice can vary for a number of reasons. Cycle-to-cycle irregularity can be associated with the inability of the vocal cords to support a periodic vibration for a defined period. Usually these types of variations are random. They are typically associated with hoarse voices.
- **MDVP:Jitter(Abs) :** Absolute jitter (/usec/) is a measurement of the pitch period's period-to period variability within the analyzed speech sample. Areas where voices break up are excluded. The voice's pitch can change for a variety of causes. Cycle-to-cycle variability may be related to the vocal cords' failure to sustain a periodic vibration for a predetermined amount of time. This kind of fluctuation typically occurs at random. Hoarse voices are frequently connected to them.
- **MDVP:RAP :** The relative average perturbation is %. a three-period smoothing factor was used to evaluate the period-to-period fluctuation of the pitch within the studied voice sample. Areas where voices break up are excluded. The inability of the vocal cords to support a periodic vibration with a known period can be linked to cycle-to-cycle inconsistency. RAP may be higher in voices that are hoarse or breathy.
- **MDVP:PPQ :** Pitch Period Perturbation Quotient /%/ - relative assessment of the pitch's period-toperiod fluctuation within the studied voice sample using a 5 period smoothing factor. The inability of the vocal folds to support a periodic vibration with a known period can be linked to cycle-to-cycle variability. Voices that are hoarse or breathy may have a higher PPQ.
- **Jitter:DDP :** Several measures of variation in fundamental frequency.
- **MDVP:Shimmer :** Shimmer Percent /%/ - /%/ Shimmer Percent - relative assessment of the peak-to-peak amplitude variability within the studied voice sample from period to period (very short term). The inability of the cords to support a periodic vibration for a specific amount of time and the existence of turbulence noise in the speech signal can both contribute to cycle-to-cycle variability of loudness.
- **MDVP:Shimmer(dB) :** Shimmer in dB /dB/ - Evaluation in dB of the period-to-period (very short-term) variability of the peak-to-peak amplitude within the analyzed voice sampleAreas where voices break up are excluded. The inability of the vocal folds to sustain a periodic vibration for a specific amount of time and the existence of turbulent noise in the voice signal can both be factors in cycle-to-cycle irregularity of loudness.

This kind of fluctuation frequently occurs at random. Voices that are breathy and harsh are frequently associated with it.

- **Shimmer:APQ3 :** The Smoothed Amplitude Perturbation Quotient is %. relative assessment of the peak-to-peak amplitude variability within the studied speech sample over the short or long term at a smoothing factor specified by the user. The smoothing factor's factory configuration is 55 periods (providing relatively long- term variability; the user can change this value as desired). Areas where voices break up are excluded. They frequently have harsh and breathy voices.
- **MDVP:APQ :** Amplitude Perturbation Quotient /%/ - relative assessment of the peak-to-peak amplitude period-to-period variability within the examined voice sample at a smoothing of 11 periods. Areas where voices break up are excluded. There are many different factors that can affect the voice's amplitude. The incapacity of the cords to maintain a periodic vibration with a known period and the existence of turbulent noise in the speech signal can be linked to cycle-to-cycle irregularity of amplitude. Hoarse and breathy voices typically have a higher APQ.
- **Shimmer:DDA :** Several measures of variation in amplitude
- **NHR :** Two Noise to Harmonic Ratio (NHR) is another useful measure. This can be routinely measured using MDVP. For a signal that can be assumed to be periodic, the signal-to-noise ratio will be equal to the harmonics-to-noise ratio.
- **HNR :** and it is this that I prefer to calculate when using Praat. Praat declares that a healthy voice phonating /a/ or /i/ should have an HNR of 20, and 40 for the phonation of the vowel /u/. Consequently, an HNR below 20 is considered to be a measure of noticeable hoarseness.
- **Status :** Health status of the subject (one) - Parkinson's, (zero) - healthy
- **RPDE :** Recurrence Period Density Entropy.
- **D2 :** It is a nonlinear dynamical complexity measures.
- **DFA :** Signal fractal scaling exponent
- **spread1, spread2, PPE :** Three nonlinear measures of fundamental frequency variation

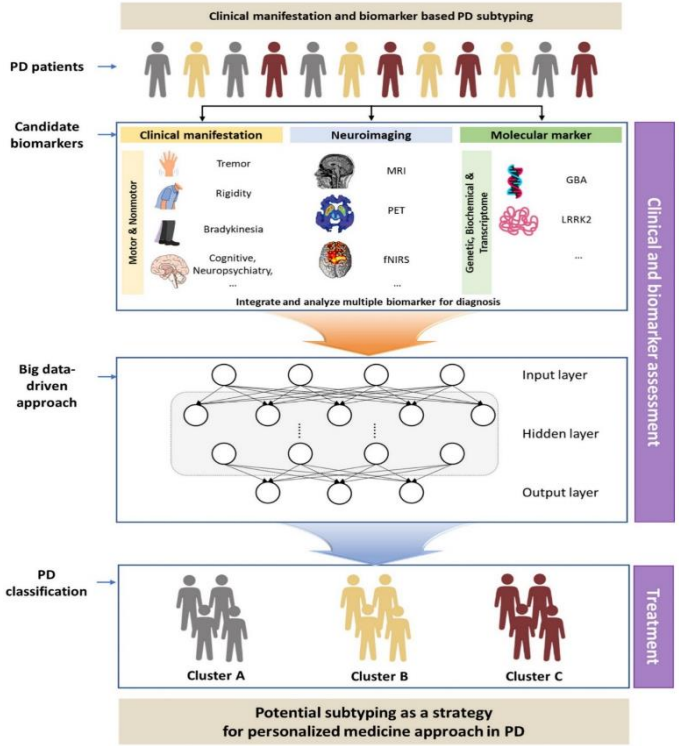

Fig 2.5.1 Clinical manifestation and biomarker based PD subtyping

The spiral dataset also collected from Kaggle repository. In training dataset contains 72 persons data (36 parkinson,36 healthy). In testing dataset contains 30 persons data (15 parkinson,15 healthy) [26].

```
├── dataset
│   ├── spiral
│   │   ├── testing
│   │   │   ├── healthy [15 entries]
│   │   │   └── parkinson [15 entries]
│   │   └── training
│   │       ├── healthy [36 entries]
│   │       └── parkinson [36 entries]
│   └── wave
│       ├── testing
│       │   ├── healthy [15 entries]
│       │   └── parkinson [15 entries]
│       └── training
│           ├── healthy [36 entries]
│           └── parkinson [36 entries]
└── detect_parkinsons.py
```

Fig 2.5.2 Hierarchy of Dataset

CHAPTER-3

SYSTEM ANALYSIS

3.1 INTRODUCTION

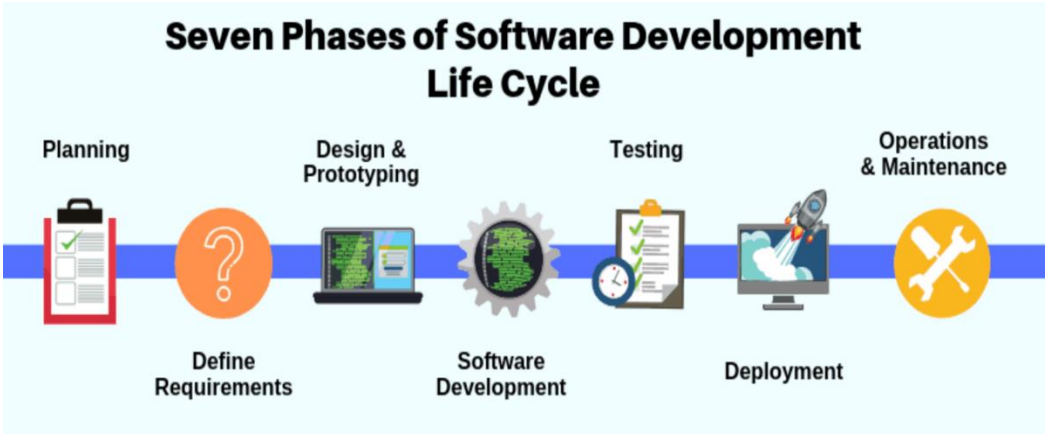

Fig 3.1 Software Development Life Cycle

System analysis is the process of examining a system or a specific aspect of a system to identify its components, functions, and interactions in order to understand how it works and to make recommendations for improvements. It involves identifying the inputs, processes, and outputs of a system, as well as the relationships between them, to gain a comprehensive understanding of the system's behavior [27]. The goal of SDLC is to minimize project risks through forward planning so that software meets customer expectations during production and beyond. This methodology outlines a series of steps that divide the software development process into tasks you can assign, complete, and measure. Requirement analysis is the most important and fundamental step in SDLC. It is performed with inputs from the customer, the sales department, market surveys and domain experts in the industry. This information is then used to plan the basic project approach and to conduct product feasibility study in the economical, operational and technical areas [27].

Planning for the quality assurance requirements and identification of the risks associated with the project is also done in the planning stage. The outcome of the technical feasibility study is to define the various technical approaches that can be followed to implement the project successfully with minimum risks. Once the requirement analysis is done the next step is to clearly define and document the product requirements and get them

approved. This is done through an SRS (Software Requirement Specification) document which consists of all the product requirements to be designed and developed during the project life cycle [28].

SRS is the reference for product architects to come out with the best architecture for the product to be developed. Based on the requirements specified in SRS, usually more than one design approach for the product architecture is proposed and documented in a DDS (Design Document Specification). This DDS is reviewed and based on various parameters as risk assessment, product robustness, design modularity, budget and time constraints, the best design approach is selected for the product [28].

A design approach clearly defines all the architectural modules of the product along with its communication and data flow representation with the external and third party modules (if any). The internal design of all the modules of the proposed architecture should be clearly defined with the minutest of the details in DDS.The actual development starts and the product is built. The programming code is generated as per DDS during this stage. If the design is performed in a detailed and organized manner, code generation can be accomplished without much hassle [27].

Developers must follow the coding guidelines defined by their organization and programming tools like compilers, interpreters, debuggers, etc. are used to generate the code. Different high level programming languages such as C, C++, Pascal, Java and PHP are used for coding. The programming language is chosen with respect to the type of software being developed [24].

System analysis can be applied to a wide range of systems, including computer software, business processes, manufacturing processes, and organizational structures. The goal of system analysis is to identify areas where the system can be optimized, streamlined, or improved in order to achieve better performance, efficiency, and effectiveness. This can involve identifying bottlenecks, redundancies, or inefficiencies within the system and developing solutions to address them [27].

System analysis can be used in a wide range of contexts, including business, engineering, healthcare, and public policy. It is often used in the development of new systems, but can also be used to improve existing systems [28].

System analysis contains the following phases :

- Information Gathering
- Analysis
- Design
- Coding
- Testing
- Deployment
- Maintenance

3.1.1 Information Gathering

In this phase, the project objectives are defined, and the requirements are gathered. The project team is assembled, and the scope of the project is established. The feasibility of the project is also assessed in terms of technical, economic, and organizational factors [17].

3.1.2 Analysis

In this phase, the requirements gathered in the previous phase are analyzed in detail to understand the functionality and features that the software should provide. This includes understanding the data needed for the algorithms, data preprocessing, and data cleaning, among other things [17].

3.1.3 Design

In this phase, the software design is created based on the requirements and analysis conducted in the previous phases. This includes creating a detailed design for the user interface, database schema, algorithms, and system architecture [17].

3.1.4 Coding

In this phase, the software development team starts coding the software based on the design specifications. The data is preprocessed, and the models are built using the Densenet and XGBoost algorithms [17].

3.1.5 Testing

In this phase, the software is tested to ensure that it meets the functional and non-functional requirements specified in the earlier phases. The software is tested for accuracy, reliability, and performance. The testing includes both unit testing and system testing [17].

3.1.6 Development

In this phase, the software is released for use in a production environment. The software is deployed on the relevant servers, and the users are provided access to the system [17].

3.1.7 Maintenance

In this phase, the software is maintained to ensure that it continues to operate as expected. This includes bug fixes, updates, and modifications to the software based on user feedback [17].

3.2 SYSTEM REQUIREMENTS

3.2.1 Software Requirements

Software : Python-version: 3.8 Any IDLE Shell

Libraries : Numpy, pandas, opencv, keras, scikit, imutils, sklearn, os etc

Operating system : windows, Linux

3.2.2 Hardware Requirements

Processor : Intel core I5

Ram : 8GB

Hard Disk : 500GB or More

3.3 SOFTWARE USED IN THE PROJECT

Python is an interpreter, high-level, general-purpose programming language. Python is simple and easy to read syntax emphasizes readability and thus reduces system maintenance costs. Python supports modules and packages, which promote system layout and code reuse. It saves space but it takes a rather higher time when its code is compiled. Indentation must be taken care while coding. Python has many inbuilt library functions that can be used easily for working with machine learning algorithms. All the necessary python libraries must be pre-installed using "pip" command. Software used in the project include following Libraries, Modules and Packages [29].

Web Application Framework or just Web Framework represents a set of libraries and modules that permits an internet application developer to write down applications without having to bother about lowlevel details such as protocols, thread management, etc. Flask may be a web application framework written in Python. It is developed by Armin Ronacher, who leads a world group of Python enthusiasts named Pocco. Flask is predicated on the Werkzeug WSGI toolkit and Jinja2 template engine. Both are Pocco projects. Python 2.6 or higher is usually required for the installation of Flask. The Flask and its dependencies work well with Python 3. Importing the flask module in the project is mandatory. An object of the Flask class is our WSGI application [30].

Libraries Used :

NumPy

NumPy is a Python library used for working with arrays. It also has functions for working in domain of linear algebra, fourier transform and matrices. NumPy was created in 2005 by Travis Oliphant. It is an open source project and you can use it freely. NumPy stands for Numerical Python.

In Alternatives of arrays in Python are lists, but they take a long time to execute. The goal of NumPy is to offer an array items that are up to 50 times quicker than conventional Python lists. The NumPy collection is referred to as ndarray, and it offers a number of supporting methods that make using ndarray very simple. In data science, where efficiency and resources are crucial, arrays are used a lot [30].

Enter the following command to install numpy : " pip install numpy "

Pandas

Python's Pandas module is used for modifying data collections.It has functions for data exploration, cleansing, analysis, and manipulation. Wes McKinney came up with the term "Pandas" in 2008, and it refers to both "Panel Data" and "Python Data Analysis". With the aid of Pandas, we can examine large data sets and draw predictions based on statistical principles. Pandas can organize disorganized data sets, making them readable and useful [30]. In data science, relevant info is crucial. Pandas is an open-source tool designed primarily for dealing quickly and logically with relational or annotated data. It offers a range of data types and functions for working with time series and numerical data. The NumPy library serves as the foundation for this library. Pandas move quickly and has high performance and productivity for users [30].

Sk-Learn

Scikit-learn, also known as sklearn, is a popular machine learning library for Python. It provides a wide range of supervised and unsupervised learning algorithms and tools for data preprocessing, feature selection, model selection, and evaluation. Scikit-learn is built on top of other popular libraries such as NumPy, SciPy, and matplotlib, and has a simple and consistent API, making it easy to learn and use. Scikit-learn supports a variety of classification, regression, and clustering algorithms, including linear and logistic regression, decision trees, random forests, support vector machines, k-nearest neighbors, k-means clustering, and more. It also provides a range of tools for feature selection and dimensionality reduction, such as Principal Component Analysis (PCA), and for model selection and evaluation, such as cross-validation and grid search. Scikit-learn is widely used in both academia and industry and has a large and active community of developers and users who contribute to its development and documentation [30].

Sk-image

Scikit-image or skimage is an open-source image processing library for Python that is built on top of NumPy, SciPy, and matplotlib. One of the notable features of skimage is its ability to work with multidimensional images, making it suitable for processing not only 2D images but also 3D and even 4D images[30].

One of the most useful functions in skimage is `skimage.io.imwrite()`, which allows you to save an image to disk in various file formats, including PNG, JPG, TIFF, BMP, and more. This function takes two arguments: the filename to save the image to, and the NumPy array representing the image data. You can also specify various options such as the image format, the image quality, and more. Overall, the `skimage.io.imwrite()` function is a simple and convenient way to save images in your Python code, and it integrates seamlessly with other functions in skimage for image processing and analysis[30] .

CV2

CV2 is a computer vision library in Python that provides various functions and tools to perform image and video processing tasks. One of the functions provided by cv2 is the write() function, which can be used to write images and videos to disk. The write() function takes two arguments: the filename to save the image or video and the image or video itself. The image or video can be a numpy array or a cv2 image object. The write() function supports various image and video formats such as JPEG, PNG, BMP, AVI, MP4, and more[30]. Using the write() function in cv2 is straightforward. First, you need to import the cv2 module in your Python script. Then, you can read an image or video using the imread() or VideoCapture() function, respectively. Once you have processed the image or video, you can save it to disk using the write() function[30].

matplotlib.pyplot

matplotlib.pyplot is a powerful data visualization library in Python that allows users to create a wide range of plots and charts. It is built on top of the Matplotlib library, which provides low-level plotting functionality. By using pyplot, users can easily create various types of plots such as line plots, scatter plots, bar charts, histograms, and more, with just a few lines of code[30]. Pyplot provides a range of customization options for plots, including adding labels, titles, legends, and annotations to the plots. Pyplot is widely used in scientific research, data analysis, and machine learning projects, where data visualization plays a crucial role in understanding and presenting insights from data. With its rich set of features and ease of use, matplotlib.pyplot is a popular choice among data scientists and researchers for creating high-quality plots in Python[30].

CHAPTER-4

SYSTEM DESIGN

4.1 INTRODUCTION

In this project, we have used random forest and XGBoost machine learning algorithms for PD detection. Random forest is applied on spiral drawing to measure the deflection in handwriting. XGBoost is applied to the voice dataset to measure voice deflections of Parkinson disease affected people[29].

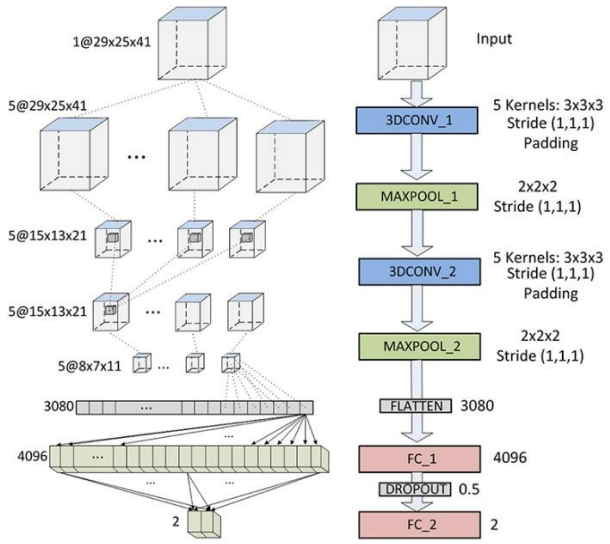

Fig 4.1 Flow of DenseNet

Fig 4.1.1 Flow Chart

4.1.1 Data Acquisition

Data set is collected from Kaggle repository. Voice data contais total 195 reords. 147 parkinson disease effected people. 48 healthy people [25]. The spiral dataset also collected from Kaggle repository. In training dataset contains 72 persons data(36 Parkinson, 36 healthy) .In training dataset contains 72 persons data (36 parkinson,36 healthy). In testing dataset contains 30 persons data (15 parkinson,15 healthy)[26].

4.1.2 Data Pre-Processing

The main aim of this step is to study and understand the nature of data that was acquired in the previous step and also to know the quality of data. A real-world data generally contains noises, missing values, and maybe in an unusable format that cannot be directly used for machine learning models. Data pre-processing is a required task for cleaning the data and making it suitable for a machine learning model which also increases the accuracy and efficiency of a machine learning model. Identifying duplicates in the dataset and removing them is also done in this step. In this project, Data processing means apply min max scalers which reduces all values of features to a value between -1 and 1. All spiral images are reshaped to desired size [17].

4.1.3 Feature Selection

Features are selected based on ranking method. Voice features dataset usually suffers from curse of dimensionality problem. Increases the time constrains and degrades classification accuracy. Hence, it is important for reducing dimensionality by selecting the relevant features. Ranking based feature selection technique has been implemented using XGBoost [18].

4.1.4 Classification

XGBoost classification algorithm is the classifier used to classify disease based on the speech and the spiral designs as the input [19]. The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups.

4.2 METHODOLOGY

4.2.1 Detecting PD using Voice Deflections

As Voice deflection has been used as a measure to detect PD because it is the most ubiquitous symptom that has been observed in more people. Different algorithms are used to measure the voice's deflection in order to identify the sickness based on the formulas. Dataset for voices was collected from the Kaggle website. A separate speech measurement that aids in the diagnosis of PD is mentioned in each column. Status indicates 0/1. 1 indicates disease and 0 indicate normal conditions [20].

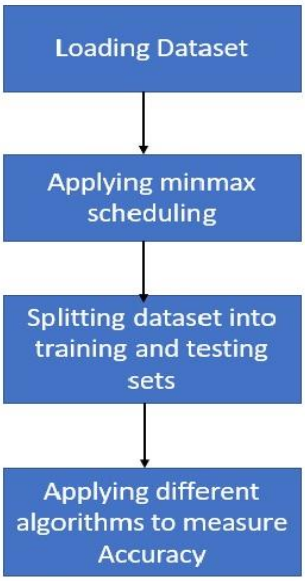

Fig 4.2.1 Flow of PD in Voice detection

- **Loading dataset:** The is first step in voice deflection analysis. First, we have load the data set downloaded from Kaggle repository to our working directory [19].
- **Applying min max Scheduler:** This is immediate step have to be after loading the dataset. Min max scheduler reduces each value of dataset between -1 to 1. This is prepoceesing technique [19].
- **Splitting:** Entire dataset was splitted into training and testing dataset. 60% of dataset is used to train the model. 40% of dataset is used test the model [19].

- **Applying machine learning algorithms:** This is the last step. Multiple algorithms were applied on the training and test dataset and accuracy was measured for best algorithm [19].

a) KNN (K Nearest Neighbor)

The simplest machine learning method that belongs to the realm of supervised learning is K Nearest Neighbor. It is quite simple to use and comprehend this method. The typical names for KNN are lazy learner algorithm and non-parametric algorithm. The main objective of this approach is classification. A complicated procedure cannot be handled by this algorithm [30].

b) Logistic Regression

The most used supervised learning method is logistic regression. The categorical data are subjected to this. On the basis of a collection of independent variables, it is used to forecast the results of categorical data. YES or 1 indicates the presence of disease, while NO or 0 indicates its absence. It is mostly used for output of the YES/NO variety based on a single attribute [30].

c) Support Vector Machine

Problems involving classification and regression may both be solved using the support vector machine approach. It is a machine-learning technique. It gives the results accurately when compared with other techniques. In realtime, there is a chance of increasing data. So, using this model is not a great decision [30].

d) XGBoost

The term "extreme gradient boosting" is "XGBoost." XGBoost is now the most popular machinelearning algorithm. When XGBoost is put up against other algorithms from the existing model, it consistently produces excellent accuracy. This results in an accuracy of 93.4%, which is very impressive. Based on the results of the decision tree, the algorithm operates. A boosting method called XGBoost builds decision trees based on the weights as shown in fig.7 describes the methodology of the XGBoost classifier [30].

4.2.2 Detecting PD using Spiral Drawings

In this, the person is asked to draw spiral drawings. These spiral models are tested with a previously trained model to detect PD. Sprial drawing data set has taken from Kaggle repository. The input image should be in the form of the fig.8. shown below. The images, collected from several public data sets A single data set was created by combining the Hand PD data set, Parkinson's Drawings data set, and the improved spiral test employing a digitized graphics tablet for monitoring PD.

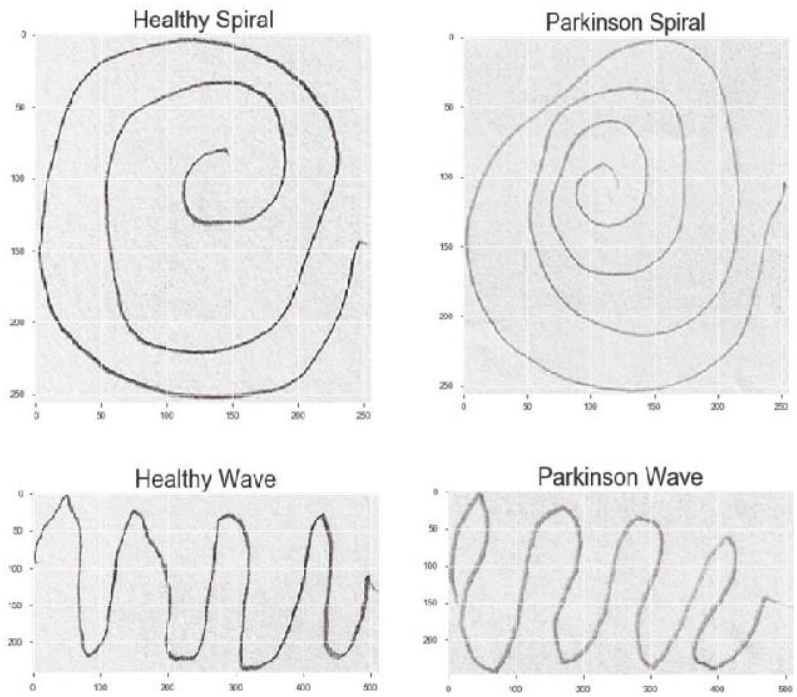

Fig 4.2.2 Spiral and Wave Drawings

For feature extraction from spiral drawings, we have used HOG technique

Quantify the image using a HOG Descriptor:

HOG is a structural descriptor that will capture and quantify changes in local gradient in the input image. HOG will naturally be able to quantify how the directions of a both spirals and waves change. And furthermore, HOG will be able to capture if these drawings have more of a "shake" to them, as we might expect from a Parkinson's patient. The resultant feature vector will then be used to train the classifier.

To quantify an image using a Histogram of Oriented Gradients (HOG) descriptor, we need to follow these steps [30].

- **Pre-Processing :** First, we need to preprocess the image by converting it to grayscale and applying any necessary normalization or filtering.
- **Cell size and block size :** Next, we need to decide on the cell size and block size to be used in the HOG computation. The cell size defines the spatial resolution of the HOG feature, while the block size defines the region over which normalization is performed.
- **Gradient computation :** We compute the gradients of the image in the x and y directions using a derivative filter (such as Sobel). This gives us the gradient magnitudes and orientations at each pixel.
- **Orientation binning :** We then quantize the gradient orientations into a set of predefined bins (usually 9) and accumulate the gradient magnitudes into each bin.
- **Block normalization :** Finally, we normalize the HOG descriptors in each block by dividing the block into overlapping cells and applying a normalization function (such as L2-norm). This helps to make the descriptor more robust to lighting and contrast changes.
- **Descriptor Calculation :** The final HOG descriptor is calculated by concatenating the normalized histograms from all of the cells within a block, and then moving the block across the entire image to compute the descriptor for the entire image.

The output of this process is a feature vector that represents the HOG descriptor of the input image. This feature vector can be used as input to a machine learning algorithm for tasks such as object detection and recognition. The resulting HOG descriptor can be used for various tasks such as object detection, pedestrian detection, or facial recognition. The descriptor captures the local gradient information in the image and is invariant to illumination and contrast changes, making it a useful feature for computer vision applications. The fig.9 describes the steps involved in spiral dataset analysis [31].

Loading Dataset : The is first step in voice deflection analysis. First, we have load the data set downloaded from Kaggle repository to our working directory [31].

Resizing the images : As all the spiral drawings cannot be in same size, in order to obtain accurate results we have to resize the spiral images to a certain size. Following are the steps have to followed to resize the images [31].

1) Load the image into memory using an image processing library, such as Pillow or OpenCV.

2) Determine the desired size of the resized image. This can be done either by specifying the width and height in pixels or by specifying a scaling factor that indicates how much the image should be scaled up or down.

3) Use an image resizing function to resize the image. The exact function will depend on the library you are using, but typical options include resize() in Pillow or resize() in OpenCV. These functions will resample the image pixels to the new size while preserving the aspect ratio.

4) Save the resized image to disk in a suitable file format, such as JPEG or PNG.

Labelling : Spiral images have to be labelled as healthy or Parkinson. This is used to understand the output.

Generating NumPy array : Features and label are appended to the data and labels lists respectively. Finally, data and labels are converted to NumPy arrays and returned conveniently in a tuple.

CHAPTER-5

SYSTEM TESTING

5.1 OVERVIEW OF TESTING

System testing is a type of software testing that evaluates the entire system or software application as a whole rather than testing its individual components or modules. It is a crucial step in the software development life cycle (SDLC) that ensures that the system or software meets the required specifications and performs its intended functions correctly and efficiently [31]. System Testing is performed on a complete integrated system to evaluate the compliance of the system with the corresponding requirements. In system testing, integration testing passed components are taken as input. System Testing is carried out on the whole system in the context of either system requirement specifications or functional requirement specifications or in the context of both. System testing tests the design and behaviour of the system and also the expectations of the customer. It is performed to test the system beyond the bounds mentioned in the SRS. System Testing is basically performed by a testing team that is independent of the development team that helps to test the quality of the system impartial. It has both functional and non-functional testing [29].

System Testing is a black-box testing. System Testing is performed after the integration testing and before the acceptance testing. System testing involves testing the software in a real-world environment, simulating end-users' interactions with the system, and verifying that the system meets the requirements and business needs. This includes testing of functional requirements, performance, usability, security, reliability, compatibility, and other aspects of the system [28].

The system testing process typically includes the following steps:

1. **Test planning:** Defining test objectives, test scenarios, and test cases to be executed.
2. **Test execution:** Running the tests and recording the results.
3. **Defect management:** Tracking and reporting defects found during testing.
4. **Test closure:** Analyzing the test results, generating reports, and obtaining approvals to release the software.

5.2 TYPES OF TESTS

5.2.1 Unit Testing

Unit testing is a software testing method in which individual units or components of a software system are tested in isolation from the rest of the system. The goal of unit testing is to verify that each unit or component of the software system is functioning as expected. In unit testing, the unit being tested is typically a function, method, or small piece of code. The unit is tested using test cases, which are input values and expected output values that are used to verify that the unit is functioning correctly. Unit testing is an important part of the software development process, as it helps to identify defects and errors early in the development cycle. By catching defects early, it is often easier and less expensive to fix them. Unit testing can also help to ensure that changes to the code do not introduce new defects. Unit testing is often automated using testing frameworks, which can execute tests automatically and provide feedback on test results. Popular testing frameworks for unit testing include JUnit for Java, NUnit for .NET, and PHPUnit for PHP [25].

5.2.2 Integration Testing

Integration testing is a software testing technique that tests the interaction between different modules or components of a software system. The purpose of integration testing is to identify any issues or defects that may arise when these different modules or components are combined and to ensure that they work together as expected. Integration testing can be performed at different levels of a software system, including unit integration testing, subsystem integration testing, and system integration testing. The goal is to identify any issues that may arise due to the integration of these different levels of the software. During integration testing, various testing techniques such as top-down, bottom-up, and incremental testing can be used to verify the interactions between the different components or modules. The testing can be done using either manual or automated testing methods, depending on the complexity of the system being tested [25].

CHAPTER-6

RESULTS

6.1 EXPERIMENTAL RESULTS OF VOICE DEFLECTIONS

We have used several measures to evaluate the predictive results. These measures are Confusion Matrix, Accuracy, Precision, Recall or Sensitivity, Specificity, F1-score.

For machine learning processes, a classification model's or algorithm's performance is represented using the confusion matrix, which can be either a chart or a table. The confusion matrix is also called as Error matrix. It is a table that is often used to describe the performance of a classification method on a set of test data for which actual value are known. Each column of the matrix represents the instances in a predicted class [29]. The correlation matrix is represented as given in the blow fig.11.

	Positive	Negative
Positive	TP	FN
Negative	FP	TN

Fig 6.1 Confusion Matrix

Confusion matrix is the useful tool for determining which tasks a machine learning system executes correctly and incorrectly. It includes both the systems predicted and true values when constructing a confusion matrix. Each predicted values has a related row and each true values has a related column [18].

A confusion matrix presents a table layout of the different outcomes of the prediction and results of a classification problem and helps visualize its outcomes. It plots a table of all the predicted and actual values of a classifier. The Confusion matrix for the voice data

analysis using the XGBoost Algorithm is shown in the below figure 12. The fig.12 describes that the predicted healthy persons are 22 and the PD persons are 6 in the dataset.

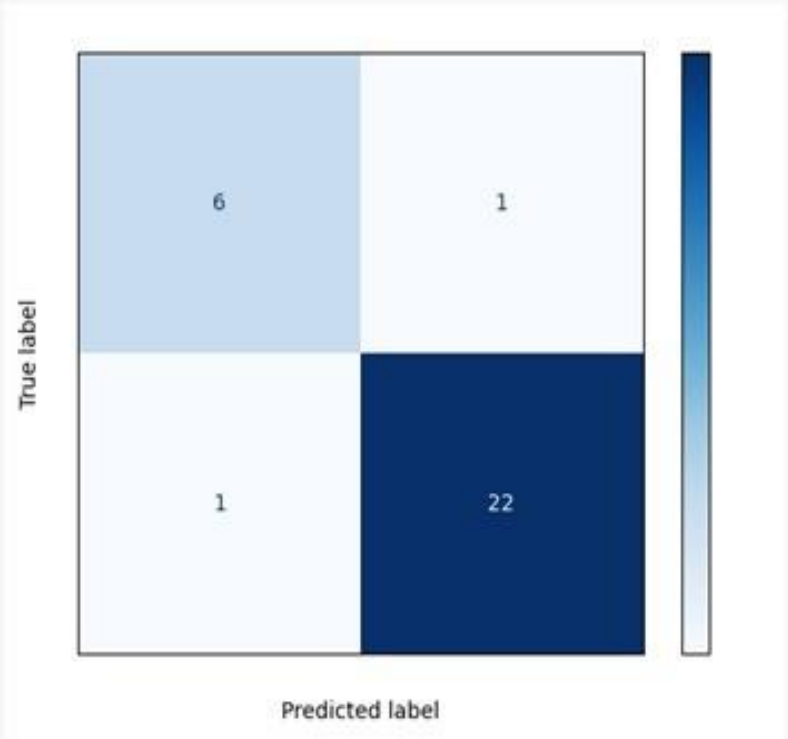

Fig 6.1.1 Confusion matrix of Speech input

In the field of machine learning and specifically the problem of statistical classification, a confusion matrix, also known as error matrix,[11] is a specific table layout that allows visualization of the performance of an algorithm, typically a supervised learning one; in unsupervised learning it is usually called a matching matrix.

It is a special kind of contingency table, with two dimensions ("actual" and "predicted"), and identical sets of "classes" in both dimensions (each combination of dimension and class is a variable in the contingency table). The different classification metrics like accuracy, sensitivity, specificity, precision, recall and recall of the proposed algorithm are measured and is shown in the below fig.13. The accuracy is measuredusing the different machine leaning algorithms with the speech data as input.

Model	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F-Score (%)
NB	83	89	82	85	84
KNN	84	87	85	82	85
SVM	86	90	87	86	87
ADA	84	88	85	83	85
GB	76	82	77	75	77
BAG	84	88	85	83	85
RF	83	87	84	82	84
XGB	**93.3**	**91.5**	**91.5**	**83.3**	**91.5**

Fig 6.1.2 Table I : Comparison of different ML algorithms with XGBoost algorithm

The testing of the data is done by using the data from the dataset and give that data as input. Fromthe data except the status column has given to the input which includes 22 parameters. The numpy array has given to the scaler transform which will minimize and reshape all the values in the range (1,-1). Afterscaling the input the model will predict the disease. If the output is '0' it says that the person does nothave the PD, else it says the person is having the PD [20].

```
In [3]: df = pd.read_csv("parkinsons.data")
        df.tail()
```

Out[3]:

	name	MDVP:Fo(Hz)	MDVP:Fhi(Hz)	MDVP:Flo(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	...	Shi
190	phon_R01_S50_2	174.188	230.978	94.261	0.00459	0.00003	0.00263	0.00259	0.00790	0.04087	...	
191	phon_R01_S50_3	209.516	253.017	89.488	0.00564	0.00003	0.00331	0.00292	0.00994	0.02751	...	
192	phon_R01_S50_4	174.688	240.005	74.287	0.01360	0.00008	0.00624	0.00564	0.01873	0.02308	...	
193	phon_R01_S50_5	198.764	396.961	74.904	0.00740	0.00004	0.00370	0.00390	0.01109	0.02296	...	
194	phon_R01_S50_6	214.289	260.277	77.973	0.00567	0.00003	0.00295	0.00317	0.00885	0.01884	...	

5 rows × 24 columns

```
In [4]: df.describe()
```

Out[4]:

	MDVP:Fo(Hz)	MDVP:Fhi(Hz)	MDVP:Flo(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer(dB)	.
count	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	.
mean	154.228641	197.104918	116.324631	0.006220	0.000044	0.003306	0.003446	0.009920	0.029709	0.282251	.
std	41.390065	91.491548	43.521413	0.004848	0.000035	0.002968	0.002759	0.008903	0.018857	0.194877	.
min	88.333000	102.145000	65.476000	0.001680	0.000007	0.000680	0.000920	0.002040	0.009540	0.085000	.
25%	117.572000	134.862500	84.291000	0.003460	0.000020	0.001660	0.001860	0.004985	0.016505	0.148500	.
50%	148.790000	175.829000	104.315000	0.004940	0.000030	0.002500	0.002690	0.007490	0.022970	0.221000	.
75%	182.769000	224.205500	140.018500	0.007365	0.000060	0.003835	0.003955	0.011505	0.037885	0.350000	.
max	260.105000	592.030000	239.170000	0.033160	0.000260	0.021440	0.019580	0.064330	0.119080	1.302000	.

8 rows × 23 columns

Fig 6.1.3 Features of the Voice Dataset

```
In [15]: from xgboost import XGBClassifier
         from sklearn.metrics import accuracy_score

In [16]: model = XGBClassifier()
         model.fit(x_train, y_train)

Out[16]: XGBClassifier(base_score=None, booster=None, callbacks=None,
                       colsample_bylevel=None, colsample_bynode=None,
                       colsample_bytree=None, early_stopping_rounds=None,
                       enable_categorical=False, eval_metric=None, feature_types=None,
                       gamma=None, gpu_id=None, grow_policy=None, importance_type=None,
                       interaction_constraints=None, learning_rate=None, max_bin=None,
                       max_cat_threshold=None, max_cat_to_onehot=None,
                       max_delta_step=None, max_depth=None, max_leaves=None,
                       min_child_weight=None, missing=nan, monotone_constraints=None,
                       n_estimators=100, n_jobs=None, num_parallel_tree=None,
                       predictor=None, random_state=None, ...)

In [22]: y_prediction = model.predict(x_test)
         print("Accuracy Score is", accuracy_score(y_test, y_prediction) * 100)

         Accuracy Score is 100.0
```

Fig 6.1.4 Accuracy rate in Voice Dataset

6.2 EXPERIMENTAL RESULTS OF SPIRAL DESIGNS

The different classification metrics like accuracy, sensitivity, specificity, precision, recall and recall of the proposed algorithm are measured. These metrics are measured using the different machine leaning algorithms with the spiral drawings data as input which will give the predictions of the given data. Here below we can see the Model accuracy and Model loss graphs of the wave and spiral drawings.

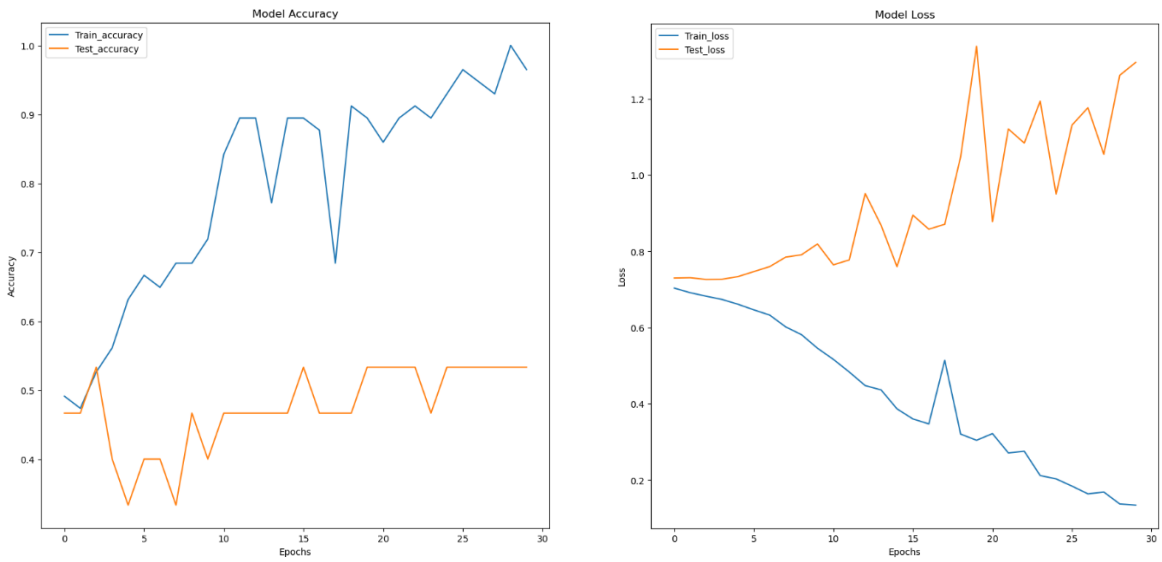

Fig 6.2.1 Model Accuracy and Model Loss in Wave Drawings

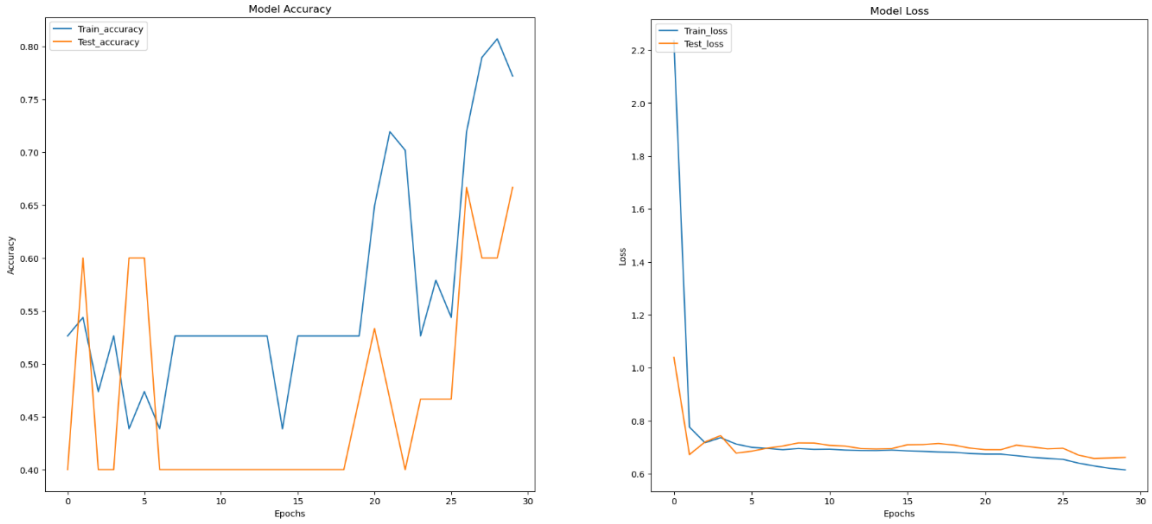

Fig 6.2.2 Model Accuracy and Model Loss in Spiral Drawings

The comparison of the different machine learning algorithms with the present algorithm to detectthe PD based on the handwritten drawings is shown in the below Table-II. Table II contains the differentparameters which are used to check the present used machine learning algorithm is better or not. The comparisons made from the previously used algorithms with the proposed algorithms is shown in the below table II. From the figure and the table II, the proposed model is having the better accuracy which is 86.67% and it is considerable compared to the other previously used algorithms.

Model	**Accuracy (%)**	**Specificity (%)**	**F-Measure (%)**	**Sensitivity (%)**
GNB	65.46	81.94	63.8	48.49
DT	62.03	65.27	70.3	58.78
LDA	68.33	81.94	68.7	54.72
KNN	64.63	70.83	70.6	58.44
SVM (Lin)	51.40	41.66	69.7	61.44
SVM (RBF)	64.54	51.38	81.9	77.70
RF	**86.67**	**93.33**	**85.71**	**80.00**

Fig 6.2.3 Table II : Comparison of different ML algorithms with Random Forest algorithm

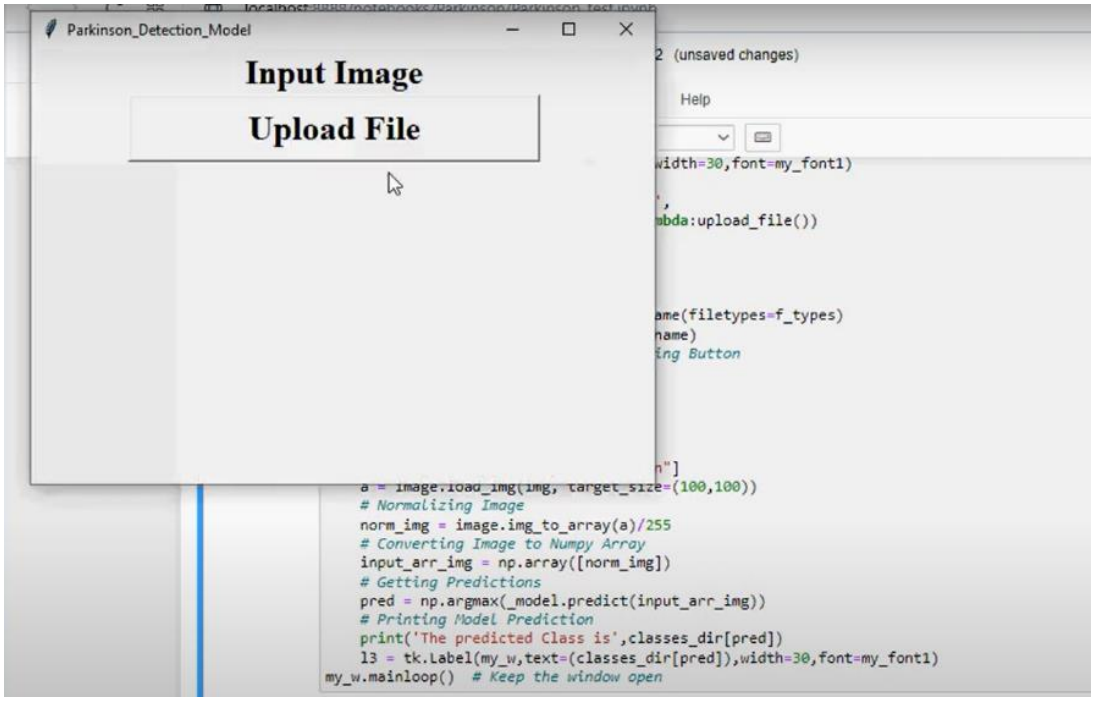

Fig 6.2.4 New window to upload the image

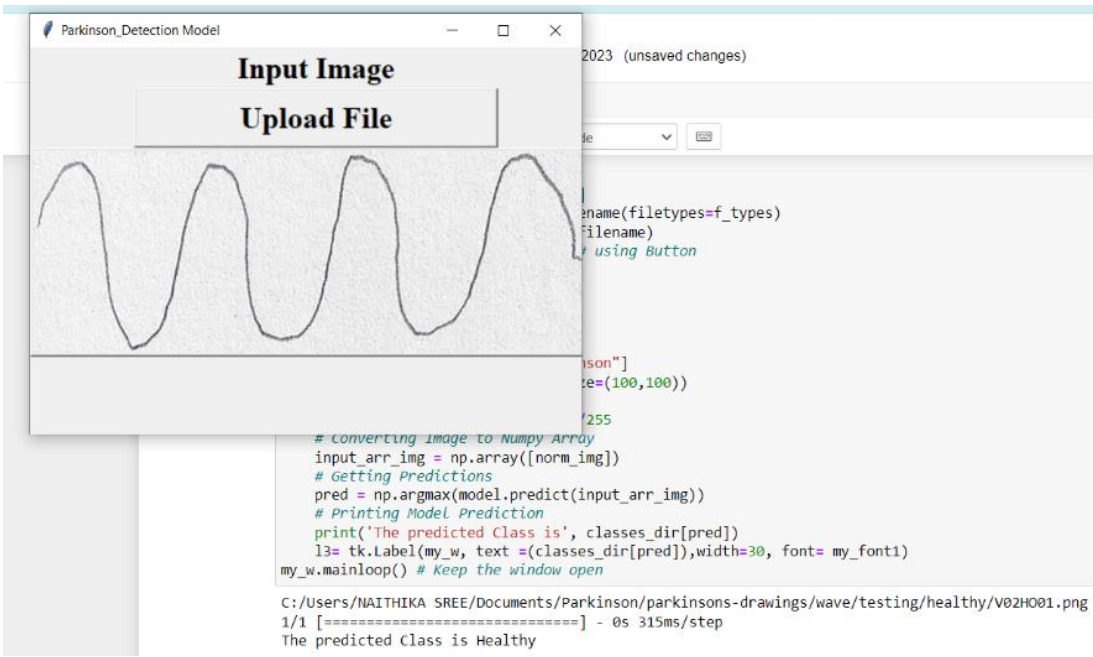

Fig 6.2.5 Predicted the image as a Healthy person's drawing

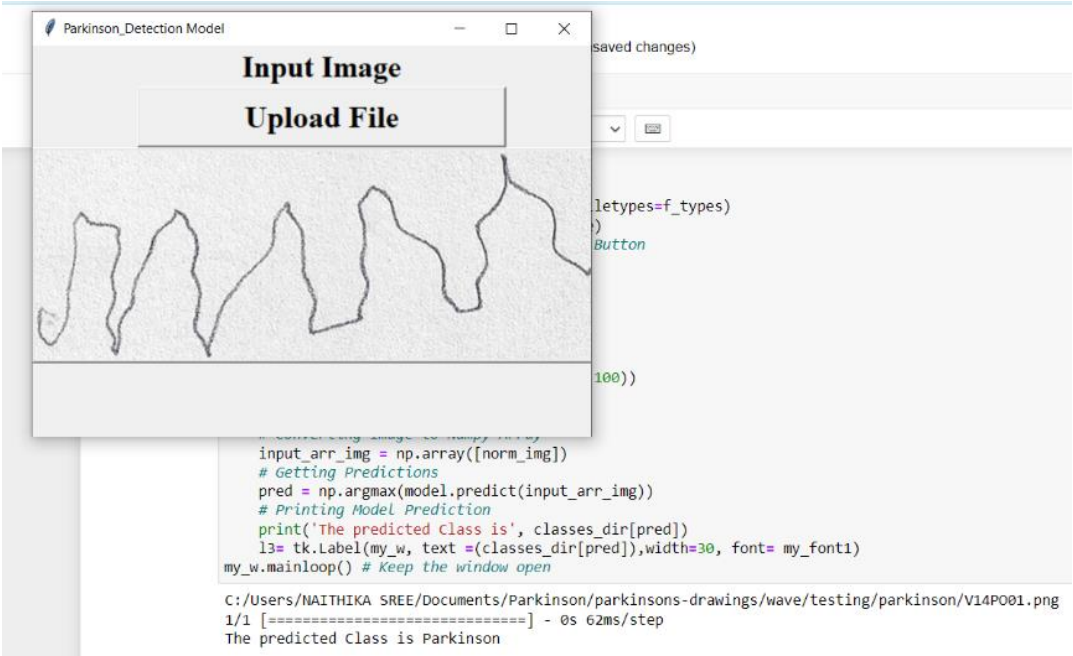

Fig 6.2.6 Predicted the image as PD affected person's drawing

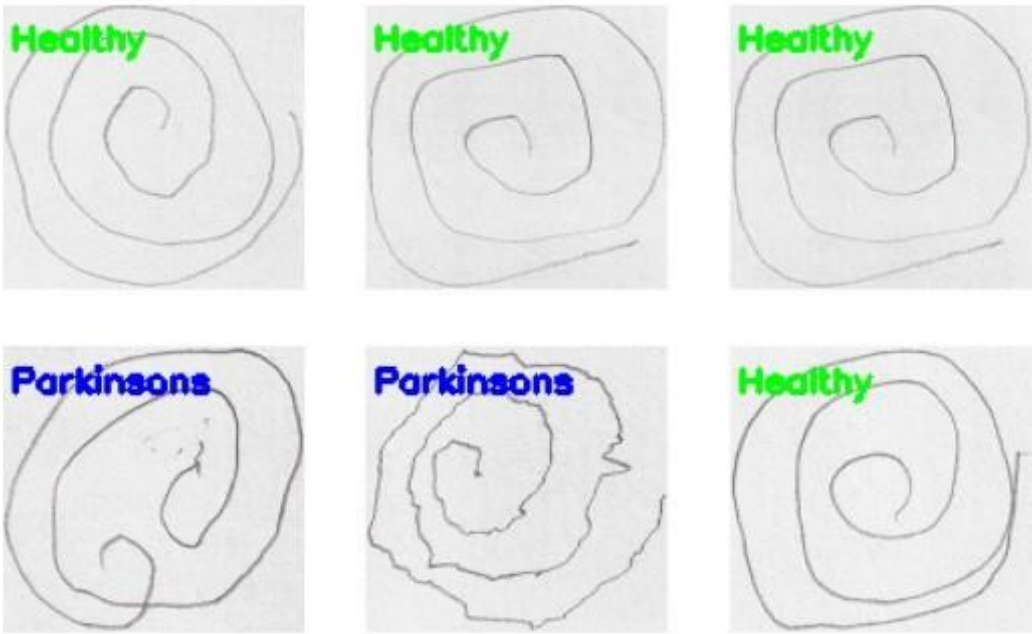

Fig 6.2.7 Healthy and PD affected spiral drawings

CHAPTER-7

CONCLUSION

Over the past 30 years, our knowledge of PD has significantly increased, with the recognition of a prodromal phase, risk, and clinical markers, along with genetic variations in PD. A trial design symposium, summarizes and highlights the progress that has been made and which, as a result, has allowed the PD field to begin contemplating identification trials. As per the above elaborated review on the disease of Parkinson's, it clearly construed that the detection of this disease needs so much effort. The detection process using machine learning technology helps the process of detection easy for humans in less cost, such that treatment and support can be provided to patients as soon as possible. Here, we presented included studies in a high-level summary, providing access to information including (a) machine learning methods that have been used in the diagnosis of PD and associated outcomes, (b) types of clinical, behavioural and biometric data that could be used for rendering more accurate diagnoses, (c) potential biomarkers for assisting clinical decision making, and (d) other highly relevant information, including databases that could be used to enlarge and enrich smaller datasets. In summary, realization of machine learning-assisted diagnosis of PD yields high potential for a more systematic clinical decision-making system, while adaptation of novel biomarkers may give rise to easier access to PD diagnosis at an earlier stage. Hence, the best efforts to be kept on finding the mechanism for the detection of the disease. The proposed project is to find Parkinson`s disease using voice data sets for testing by XGBoost algorithm and examine handwritten drawings by using random forest algorithm. The analysis of the voice and drawings data gives the accurate performance in detecting the PD. The accuracy of the proposed project for speech data is 100% and for handwritten spiral drawings 83.333%.

CHAPTER-8

FUTURE SCOPE

In the present decade of accelerated advances in Medical Sciences, most studies fail to lay focus on ageing diseases. These are diseases that display their symptoms at a much advanced stage and makes a complete recovery almost improbable. Parkinson's disease (PD) is the second most commonly diagnosed neurodegenerative disorder of the brain. One could argue, that it is almost incurable and inflicts a lot of pain on the patients. All these make it quite clear that there is an oncoming need for efficient, dependable and expandable diagnosis of Parkinson's disease. Managing PD in day-to-day life is very challenging for an individual. Therefore, a good screening procedure will be beneficial, especially in circumstances where a physician's treatment is not necessary. Thus, for the diagnosis of PD, ML algorithms were evaluated. The main aim of this was to identify existing ML-based research to diagnose PD in terms of handwritten patterns, voice attributes, and to determine the most appropriate technique to diagnose the PD with an accuracy rate. This report also mentions the developments in neural networks and related learning systems, which provide valuable insights and guidelines for future progress. In future work, we can focus on different techniques to predict the Parkinson disease using different datasets. In this research, we using binary attribute (1- diseased patients, 0-non-diseased patients) for patient's classification. In the future we will use different types of attributes for the classification of patients and also identify the different stages of Parkinson's disease which really helps especially for the families who cannot afford the expensive MRI scans. In the future as this will be more advanced where we can use the position and the movements of the body to identify the disease with more accuracy. We observed that there is still a lot of work that has to be performed in the future.

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ANNEXURE

PARKINSON'S DISEASE DETECTION USING VOICE DEFLECTIONS

```
import numpy as np

import pandas as pd

import os, sys

df = pd.read_csv("parkinsons.data")

df.tail()

df.describe()

df.info()

df.shape

features = df.loc[:, df.columns != 'status'].values[:, 1:]

labels = df.loc[:, 'status'].values

from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler((-1, 1))

X = scaler.fit_transform(features)

y = labels

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test=train_test_split(X, y, test_size=0.30)

from xgboost import XGBClassifier

from sklearn.metrics import accuracy_score

model = XGBClassifier()

model.fit(x_train, y_train)

y_prediction = model.predict(x_test)
```

```
print("Accuracy Score is", accuracy_score(y_test, y_prediction) * 100)
```

PARKINSON'S DISEASE DETECTION USING SPIRAL DRAWINGS

Code 1 :

```
import numpy as np

import pandas as pd

import os

import matplotlib.pyplot as plt

import tensorflow as tf

from tensorflow.keras.utils import to_categorical

from tensorflow.keras.preprocessing.image import load_img, img_to_array

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Dense, Dropout, Flatten, Conv2D, MaxPooling2D

from tensorflow.keras.callbacks import EarlyStopping

from tensorflow.python.ops.numpy_ops import np_utils

from sklearn.metrics import classification_report, log_loss, accuracy_score

from sklearn.model_selection import train_test_split

from keras.preprocessing.image import ImageDataGenerator

from tensorflow.keras.optimizers import Adam

import warnings

warnings.filterwarnings("ignore")

import numpy as np

import cv2
```

```
from tensorflow.keras.layers import Input, Conv2D, BatchNormalization, Dropout,
Flatten, Dense, MaxPool2D

from tensorflow.keras.models import Model, Sequential

from tensorflow.keras.initializers import glorot_uniform

from tensorflow.keras.optimizers import Adam, SGD

import matplotlib.pyplot as plt

from sklearn.metrics import classification_report

from tensorflow.keras.preprocessing.image import ImageDataGenerator

from sklearn.model_selection import train_test_split

from tensorflow.keras.utils import to_categorical

from sklearn.preprocessing import LabelEncoder

from sklearn.metrics import classification_report, confusion_matrix

import seaborn as sns

from tensorflow.keras.regularizers import l2

dir_sp_train = './parkinsons-drawings/spiral/training'

dir_sp_test = './parkinsons-drawings/spiral/testing'

dir_wv_train = './parkinsons-drawings/wave/training'

dir_wv_test = './parkinsons-drawings/wave/testing'

# Says how many files are present in that particular file/folder

Name=[]

for file in os.listdir(dir_sp_train):

Name+=[file]

print(Name)

N=[]
```

```
# To access each item of the sequence with the help of index
for i in range(len(Name)):
N+=[i]
normal_mapping=dict(zip(Name,N))
reverse_mapping=dict(zip(N,Name))
# Tranform maps string value from one to another
def mapper(value):
return reverse_mapping[value]
dataset_sp=[]
count=0
for file in os.listdir(dir_sp_train):
# Combine paths names into one complete path
path=os.path.join(dir_sp_train,file)
for im in os.listdir(path):
image=load_img(os.path.join(path,im),grayscale=False, color_mode='rgb',
target_size=(100,100))
image=img_to_array(image)
image=image/255.0
# Adds single item to certain collection types
dataset_sp.append([image,count])
count=count+1
testset_sp=[]
count=0
for file in os.listdir(dir_sp_test):
```

```
path=os.path.join(dir_sp_test,file)

for im in os.listdir(path):

image=load_img(os.path.join(path,im),grayscale=False,color_mode='rgb',
target_size=(100,100))

image=img_to_array(image)

image=image/255.0

testset_sp.append([image,count])

count=count+1

dataset_wv=[]

count=0

for file in os.listdir(dir_wv_train):

path=os.path.join(dir_wv_train,file)

for im in os.listdir(path):

image=load_img(os.path.join(path,im), grayscale=False, color_mode='rgb',
target_size=(100,100))

image=img_to_array(image)

image=image/255.0

dataset_wv.append([image,count])

count=count+1

testset_wv=[]

count=0

for file in os.listdir(dir_wv_test):

path=os.path.join(dir_wv_test,file)

for im in os.listdir(path):
```

```
image=load_img(os.path.join(path,im), grayscale=False, color_mode='rgb',
target_size=(100,100))

image=img_to_array(image)

image=image/255.0

testset_wv.append([image,count])

count=count+1

# Takes in iterables as arguments and return iterators

data_sp,labels_sp0=zip(*dataset_sp)

test_sp,tlabels_sp0=zip(*testset_sp)

data_wv,labels_wv0=zip(*dataset_wv)

test_wv,tlabels_wv0=zip(*testset_wv)

# Converts an integer to binary matrix

labels_sp1=to_categorical(labels_sp0)

data_sp=np.array(data_sp)

labels_sp=np.array(labels_sp1)

tlabels_sp1=to_categorical(labels_sp0)

test_sp=np.array(test_sp)

tlabels_sp=np.array(tlabels_sp1)

labels_wv1=to_categorical(labels_wv0)

data_wv=np.array(data_wv)

labels_wv=np.array(labels_wv1)

tlabels_wv1=to_categorical(labels_wv0)

test_wv=np.array(test_wv)

tlabels_wv=np.array(tlabels_wv1)
```

```
trainx_sp,testx_sp,trainy_sp,testy_sp=train_test_split(data_sp,labels_sp,test_size=0.2,rand
om_state=44)

trainx_wv,testx_wv,trainy_wv,testy_wv=train_test_split(data_wv,labels_wv,test_size=0.2,r
andom_state=44)

print(trainx_sp.shape)

print(testx_sp.shape)

print(trainy_sp.shape)

print(testy_sp.shape)

print(" CNN Model Build")

# To create models layer by layer

model= Sequential()

# Convolution: Deep learning algorithm specially designed for working images and videos

model.add(Conv2D(filters=32, kernel_size=(5,5),activation="relu",
input_shape=(100,100,3))) #applys conv to images

model.add(MaxPool2D(pool_size=(2,2))) #adjusts the image's pixels

# 2nd Convolutional layer

model.add(Conv2D(filters=64, kernel_size=(5,5),activation="relu"))

model.add(MaxPool2D(pool_size=(2,2)))

model.add(Flatten()) #returns the copy of array

# 1st Fully Connected Layer

model.add(Dense(units=64,activation="relu"))

# Add Output Layer

model.add(Dense(units=2,activation="softmax"))#classification of tasks

print ("Model Summary")

model.summary() # Print Summary of the Model
```

```
model.compile(loss='categorical_crossentropy', optimizer= Adam(lr=0.001),
metrics=['accuracy'])

# Timer

epoch =30

# Samples processed before the model is updated

batch_size=32

print("Start Training", '\n')

hist_0= model.fit(trainx_sp,trainy_sp, batch_size=batch_size, epochs=epoch,
validation_data=(testx_sp, testy_sp))#provides fit statistics

hist_1= model.fit(trainx_wv,trainy_wv, batch_size=batch_size, epochs=epoch,
validation_data=(testx_wv, testy_wv))

print("Training End", '\n')

model.save ("CNN_Model.h5")

figure=plt.figure(figsize=(10,10))

plt.plot(hist_0.history['accuracy'],label='Train_accuracy')

plt.plot(hist_0.history['val_accuracy'],label='Test_accuracy')

plt.title('Model Accuracy')

plt.xlabel('Epochs')

plt.ylabel('Accuracy')

plt.legend(loc="upper left")

plt.show()

figure2=plt.figure(figsize=(10,10))

plt.plot(hist_0.history['loss'],label='Train_loss')

plt.plot(hist_0.history['val_loss'],label='Test_loss')

plt.title('Model Loss')
```

```
plt.xlabel('Epochs')

plt.ylabel('Loss')

plt.legend(loc="upper left")

plt.show()

figure = plt.figure(figsize=(10,10))

plt.plot(hist_1.history['accuracy'],label='Train_accuracy')

plt.plot(hist_1.history['val_accuracy'],label='Test_accuracy')

plt.title('Model Accuracy')

plt.xlabel('Epochs')

plt.ylabel('Accuracy')

plt.legend(loc="upper left")

plt.show()

figure2 = plt.figure(figsize=(10,10))

plt.plot(hist_1.history['loss'], label='Train_loss')

plt.plot(hist_1.history['val_loss'], label='Test_loss')

plt.title('Model Loss')

plt.xlabel('Epochs')

plt.ylabel('Loss')

plt.legend(loc="upper left")

plt.show()

datagen=ImageDataGenerator(horizontal_flip=True,vertical_flip=True,rotation_range=20,
zoom_range=0.2,width_shift_range=0.2,height_shift_range=0.2,shear_range=0.1,fill_mod
e="nearest")

pretrained_model3=tf.keras.applications.DenseNet201(input_shape=(100,100,3),include_t
op=False,weights='imagenet', pooling='avg')
```

```
pretrained_model3.trainable=False

pretrained_model4=tf.keras.applications.DenseNet201(input_shape=(100,100,3),include_top=False,weights='imagenet', pooling='avg')

pretrained_model4.trainable =False

inputs3= pretrained_model3.input

x3=tf.keras.layers.Dense(128, activation='relu')(pretrained_model3.output)

outputs3=tf.keras.layers.Dense(2, activation='softmax')(x3)

model3=tf.keras.Model(inputs=inputs3, outputs=outputs3)

inputs4= pretrained_model4.input

x4=tf.keras.layers.Dense(128, activation='relu')(pretrained_model4.output)

outputs4=tf.keras.layers.Dense(2, activation='softmax')(x4)

model4=tf.keras.Model(inputs=inputs4, outputs=outputs4)

model3.compile(optimizer='adam',loss='categorical_crossentropy',metrics=['accuracy'])

model4.compile(optimizer='adam',loss='categorical_crossentropy',metrics=['accuracy'])

sp_3=model3.fit(datagen.flow(trainx_sp,trainy_sp,batch_size=32),validation_data=(testx_sp,testy_sp),epochs=100)

wv_4=model4.fit(datagen.flow(trainx_wv,trainy_wv,batch_size=32),validation_data=(testx_wv,testy_wv),epochs=100)

#Spiral

y_pred_sp=model3.predict(testx_sp)

pred_sp=np.argmax(y_pred_sp,axis=1)

ground_sp= np.argmax(testy_sp,axis=1)

print(classification_report(ground_sp,pred_sp))

# Wave

y_pred_wv=model3.predict(testx_wv)
```

```
pred_wv=np.argmax(y_pred_wv,axis=1)

ground_wv= np.argmax(testy_wv,axis=1)

print(classification_report(ground_wv,pred_wv))

get_acc3=sp_3.history['accuracy']

value_acc3=sp_3.history['val_accuracy']

get_loss3=sp_3.history['loss']

validation_loss3=sp_3.history['val_loss']

epochs3=range(len(get_acc3))

plt.plot(epochs3,get_acc3,'r', label='Accuracy of Training data')

plt.plot(epochs3,value_acc3,'b', label='Accuracy of Validation data')

plt.title('Training vs Validation Accuracy -Spiral')

plt.legend(loc=0)

plt.figure()

plt.show()

epochs3=range(len(get_loss3))

plt.plot(epochs3,get_loss3,'r', label='Loss of Training data')

plt.plot(epochs3,validation_loss3,'b', label='Loss of Validation data')

plt.title('Training vs Validation Accuracy -Spiral')

plt.legend(loc=0)

plt.figure()

plt.show()

load_img("./parkinsons-
drawings/spiral/testing/healthy/V10HE01.png",target_size=(100,100))

image= load_img("./parkinsons-
drawings/spiral/testing/healthy/V10HE01.png",target_size=(100,100))
```

```
image=img_to_array(image)

image=image/255.0

prediction_image=np.array(image)

 # New axis position in the expanded array

prediction_image=np.expand_dims(image, axis=0)

prediction=model3.predict(prediction_image)

value=np.argmax(prediction)

move_name=mapper(value)

print("Prediction is {}.".format(move_name))

load_img("./parkinsons-
drawings/wave/testing/parkinson/V03PO01.png",target_size=(100,100))

image2= load_img("./parkinsons-
drawings/wave/testing/parkinson/V03PO01.png",target_size=(100,100))

image2=img_to_array(image2)

image2=image2/255.0

prediction_image2=np.array(image)

prediction_image2=np.expand_dims(image, axis=0)

prediction2=model4.predict(prediction_image2)

value2=np.argmax(prediction2)

move_name2=mapper(value2)

print("Prediction is {}.".format(move_name2))

print(test_sp.shape)

prediction_sp=model3.predict(test_sp)

print(prediction_sp.shape)

PRED_sp=[]
```

```
for item in prediction_sp:
# Process of accessing specific element
value_sp=np.argmax(item)
PRED_sp+=[value_sp]
ANS_sp=tlabels_sp0
accuracy_sp= accuracy_score (ANS_sp, PRED_sp)
print(accuracy_sp)
print(test_wv.shape)
prediction_wv=model4.predict(test_wv)
print(prediction_wv.shape)
PRED_wv=[]
for item in prediction_wv:
value_wv=np.argmax(item)
PRED_wv+=[value_wv]
ANS_wv=tlabels_wv0
accuracy_wv= accuracy_score (ANS_wv, PRED_wv)
print(accuracy_wv)
```

Code 2 :

```
import tkinter as tk
from tkinter import filedialog
from tkinter.filedialog import askopenfile
from tkinter import messagebox
from tkinter import ttk
```

```
from tensorflow.keras.preprocessing import image

import numpy as np

import pandas as pd

import tensorflow as tf

import tensorflow.keras

from tensorflow.keras import layers

from tensorflow.keras import Model

from tensorflow.keras.models import Sequential

from tensorflow.keras.preprocessing import image

from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping

from tensorflow.keras.models import load_model, Model

from tensorflow.keras.layers import Dense, Conv2D, Flatten, MaxPool2D, Dropout

model= load_model('CNN_Model.h5')

from PIL import Image, ImageTk

model= load_model('CNN_Model.h5')

# Display new window

my_w= tk.Tk()

# Size of the window

my_w.geometry("500x350")

my_w.title('Parkinson_Detection Model')

my_font1=('times', 20, 'bold')

l1= tk.Label (my_w, text='Input Image',width =30, font=my_font1)

l1.grid(row=1, column=1)
```

```
b1= tk.Button(my_w, text='Upload File', width=20, font= my_font1, command=
lambda:upload_file())

b1.grid(row=2, column=1)

def upload_file():

global img

f_types= [('png Files','*.png')]

filename= filedialog.askopenfilename(filetypes=f_types)

img = ImageTk.PhotoImage(file= filename)

# Using Button

b2= tk.Button(my_w, image=img)

b2.grid(row=3, column=1)

print(filename)

Prediction(filename, model)

def Prediction (img, model):

classes_dir= ["Healthy", "Parkinson"]

a=image.load_img(img, target_size=(100,100))

# Normalizing Image

norm_img= image.img_to_array(a)/255

# Converting Image to Numpy Array

input_arr_img = np.array([norm_img])

# Getting Predictions

pred = np.argmax(model.predict(input_arr_img))

# Printing Model Prediction

print('The predicted Class is', classes_dir[pred])
```

```
l3= tk.Label(my_w, text =(classes_dir[pred]),width=30, font= my_font1)

# Keep the window open

my_w.mainloop()
```